



Adapting Energiewende Principles to Iraqi Context: A Phased Energy-Transition Framework Integrated With Smart Urban Planning

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Received: 05-22-2026

Revised: 08-05-2026

Accepted: 08-14-2026

Citation: Al-Salim, K. A., Hasan, A. M., & Al-Jaberi, A. A. (2026). Adapting Energiewende principles to Iraqi context: A phased energy-transition framework integrated with smart urban planning. *Chall. Sustain.*, 14(5), 884–920. <https://doi.org/10.56578/cis140503>.



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Abstract: This study reflects upon the transformability of the chosen principles of Energiewende (Germany's energy transition) and policy instruments met in the Iraqi energy and urban scenario. The study does not take it for granted that the German model is directly applicable to other countries. Instead, using secondary sources such as policy documents, institutional reports, and academic literature, this study is comparative and context-sensitive. The analysis focuses on six dimensions: technical and network readiness, financial and investment capacity, institutional and regulatory quality, community acceptance and participation, equity in energy access, and renewable energy potential and energy efficiency. Finally, the Multi-Level Perspective is used to analyze and explain how emerging niche innovations, the prevailing socio-technical regime, and the landscape pressures are connected. Results show how elements of the Energiewende can be adapted and are divided into three types: elements which need little adaptation, elements which need significant adaptation, and elements which require more advanced institutional, technical, financial and/or market conditions. Energy efficiency and distributed solar energy, along with smart metering, reduction of transmission and distribution losses, institutional capacity building, and pilot energy storage and microgrids are the most viable components for the Iraqi environment. In contrast, electricity-market liberalization, sophisticated trading mechanisms, large-scale community ownership, and ambitious decarbonization objectives call for a certain amount of contextual provision and deeper readiness of the system. Based on these results, the study proposes three stages of an adaptive pathway: System Stabilization and Enabling Foundations, Expansion and Institutional Embedding, System Integration and Structural Transformation. Movement between stages is determined by observable improvements in readiness and not by pre-determined levels or timescales. The results also showcase how the framework can be implemented at the city, neighborhood, building and infrastructure level linked by smart urban planning, which offers a spatial and institutional context to operationalize the framework. The key enabling measures are: rooftop solar systems, microgrids, energy-efficient urban development, smart metering, demand management based on data analysis, local governance, and spatial-equity safeguards. The study presents an integrated analytical framework that integrates Energiewende principles, readiness assessment, Multi-Level Perspective, and smart urban planning for situations where the performance of the grid is weak, institutions and financial resources are limited, and access to energy service is unequal. Empirical validation of this framework will be needed by engaging the various stakeholders, spatial analysis, techno-economic analysis, pilot activities, and long-term monitoring.

Keywords: Renewable energy; Smart cities; Staged Transition Model; Multi-Level Perspective; Energiewende; Grid readiness; Energy equity

1. Introduction

Urban development, population, and economic growth have led to a number of cities experiencing accelerated urban growth throughout the world. Yet this expansion has not always matched any boost in the sustainability of urban systems or in the efficiency with which urban resources were used: many cities are under pressure from environmental impacts, resource scarcity and, in some cases, the lack of some basic services (Satterthwaite, 2011; UN-Habitat, 2022). Significantly rapid urbanization and growth of economic and service activities have created a strong demand for energy and urban infrastructure, which in the majority of countries are mainly fossil fuel-based. This dependency points to increased emissions levels, air pollution, resource depletion, unfavorable supply certainty, and unequal access to regular and affordable energy, which heightens energy poverty and social inequality levels.

Thus, the urban energy transition cannot be equated with the shift from fossil fuels to renewables, but is actually a reconfiguration of the urban energy system that involves institutional, economic, and social changes as well as technological changes. In this context, smart city technologies, such as smart grids, smart meters, and data-driven demand management, can play an enabling role in the energy transition by improving system efficiency, integrating distributed energy sources, and supporting decision-making. Their effectiveness, however, depends on governance and infrastructure, data availability and accessibility, and equitable access to relevant technologies and services (Meredith, 2018). The Intergovernmental Panel on Climate Change (IPCC) Special Report on Renewable Energy Sources and Climate Change Mitigation confirms that renewable energy and improved energy efficiency are important means of reducing greenhouse gas emissions from the energy sector and improving energy-system sustainability (IPCC, 2011). However, the adoption of these technologies also depends on regulatory and financial frameworks, the capacity of energy systems to integrate renewable sources, social acceptance, and country-specific institutional conditions, in addition to their technical and economic maturity. (IPCC, 2007).

Urban agglomerations are responsible for some 70% of total greenhouse gases (GHGs) emissions, hence playing a key role in GHGs mitigation. Urgent action is needed to decrease these emissions through more robust international, national and local commitments, as well as build climate change mitigation targets into urban programming, energy planning and urban infrastructure development (OECD & UN-Habitat, 2022).

The urban model, as a dynamic and integrated model, has a need to be developed for a resource-efficient, socially just and low-carbon urban model to address these challenges of urban growth. This approach will take form in urban master planning and can be replicated by implementing measures for land-use diversity, along with renewable, self-sufficient energy sources, protection of biodiversity, and sustainable urban infrastructure (Di Liddo et al., 2025). The size and current rate of urbanization are set to accelerate on an unprecedented scale in the future, with more than 200 new urban centers being expected in Asia in the next 30 years (UNDP, 2025), and so its sustainability will rely heavily on urban land use decisions, infrastructure development, energy use efficiency, and the strength of urban institutions in guiding urban investment equitably and sustainably. These principles need, however, to be adapted to the local context in Iraq, as cities within the country face unequal access to energy services and smart technologies, are highly dependent on the use of fossil fuels, and suffer low levels of institutional capacity and finance, as well as weaknesses in the electricity grid.

In response, this study compares and examines the characteristics of the German Energiewende principles to the urban and energy situation in Iraq, and which of the principles are highly adaptable and which need to be considered for re-fitting to account for institutional, financial, technical, and social limitations. The research also introduces an adaptive phased approach with smart technologies as enablers to increase the efficiency of the electrical grid, control the demand, and integrate with distributed energy sources, but not as a final goal. Therefore, addressing the following questions is the study's aim: What can be transferred to the Iraqi context in the Energiewende concept? What transposition has to be made to adjust the rest of the elements to the Iraqi context? How can renewables and smart city technologies be integrated across a multi-phase strategy while considering grid readiness, the availability of capital to facilitate energy investments, community acceptance, and fair access to energy? It builds upon Energiewende principles, Multi-Level Perspective, and smart city technologies (including a smart grid) and creates a context-adaptive analytical framework, which has been developed based on a room-for-evolution approach tailored to the Energiewende context in the study region, featuring low grid quality and weak institutional and financial capacities.

Current research on the energy transition in Iraq reveals a variety of challenges, from institutional and regulatory obstacles to technical and network readiness, financing options, and a lack of community acceptance, all the way to digital readiness and the potential for renewable energy. A review of this literature indicates that most of the studies have been on specific aspects or a limited subset of these issues; an integrated approach that couples all these elements is less common.

Ali (2026) revealed that the institutional and regulatory hurdles to renewable energy development in Iraq are not so much related to the lack of resources or technical potential, but rather to a lack of unified legislation, overlapping mandates, opaque licensing and grid-connection processes, and investment incentives and guarantees. This is in line with the results obtained by Chenic et al. (2022), who said political will, legislation stability, the

ability to mobilize funding, and public-private partnerships are important. Taken together, these studies underline that good institutional and regulatory conditions are crucial for the feasibility of the mechanisms for energy transition, but both focus on a general or sectoral level and do not systematically connect them with the other requirements for the energy transition, such as technical, social, and digital readiness (Chenic et al., 2022).

Regarding technical and network readiness, the study by Tahir (2026) showed that microgrids and hybrid systems can be considered in an area where power supply has been found inadequate if they are appropriately adapted to the characteristic nature of the demand, reliability of the power supply, and operational and maintenance capacities. Similarly, a study mentioned by Al-Samarrai (2024) stated that Iraq has great potential to develop its energy sources with more diversity, especially in Solar Energy, but it needs to enhance the technical infrastructure, grid enhancement, and alleviate the climate for investment in order to convert this potential into practice. Despite the commonalities of decentralization and diversification of energy sources, both studies prioritize other matters; the former being more technical and operational, and the latter more natural potential, institutional, and financial readiness. This means that the technical potential alone is not enough to inform about scalability or sustainability (Al-Samarrai, 2024).

More specifically, the social aspect of distributed energy resources: another study by Hassan et al. (2024) revealed that communities must be actively involved, institutions must be trusted, communities' rights must be given to small-scale producers, and local energy ownership must be transparent. Their results showed that the energy transition is not just a technical transition but also a socio-political one, as it needs the involvement of the population and stakeholders (Hassan et al., 2024). This is supported by Al-Lami (2023) in this regard, which indicated that besides the technical feasibility of installing distributed energy systems, many other factors such as awareness, trust in technical safety, and government incentives affect the adoption process. What this intersection makes apparent is that the 'social' outcomes of these technologies are not just a function of their traditional presence or efficiency, but also their levels of acceptance, understanding of benefits, and the trust in operational and financing arrangements. Yet social literature in the field tends to stick to the themes of acceptance, or even participation in specific contexts or scenarios, and does not fully connect the themes with equitable access to energy and the spatial differences among groups/areas (Al-Lami, 2023).

Sarabdeen et al. (2024) suggested that digital technologies are not a magic bullet for the digital transformation to reach the post-carbon goal in energy consumption, but heavily rely on energy and infrastructure readiness, institutional quality and institutional capacity for financing, and the integration of digital and energy policies. This is relevant for this research because it suggests a further message for the actors who are using smart city technologies: Do not see smart city technologies as an objective in themselves or as an alternative to grid and institutional measures. Instead, it views them as enabling factors, and the effectiveness of these could depend on the readiness of the community and equal capabilities in accessing digital services. Therefore, the digital dimension is intertwined with the institutional and social dimensions and cannot be viewed without an interconnectedness with the two (Sarabdeen et al., 2024).

Overall, current research findings do show a wide agreement on the key requirements for a successful energy transition in Iraq. The latter includes the recognition of significant potential in renewable energy and energy efficiency, but essential are also grid reliability, investment capacity, as well as clear institutional and regulatory frameworks, social acceptance and engagement aspects, and equitable access. In the literature, the emphasis on these conditions is, however, different: some studies touch the subject from a legal or financial point of view, others are provided with technical aspects, and some other studies focus on social acceptance or digitalization. This suggests that the existing knowledge is still somewhat scattered and a holistic approach is lacking to examine the interplay between these dimensions for evaluating energy transition component adaptability to the Iraqi context. The research gap in this area is the lack of an analytical framework to systematically combine these dimensions, using them not necessarily to determine challenges, but to determine if energy-transition elements are highly adaptable, need significant adaptation, or are dependent on a more developed state of the system. Additionally, despite the importance of the previous studies, they have failed to combine the aspects of the energy transition in Germany and a Multi-Level Perspective with the specific grid conditions, institutional, financial, societal, and equity conditions in Iraq into a manifested phased way.

Therefore, the analytical framework of the current study is based on six dimensions, which are interlinked: technical/network readiness; financial/investment capacity; institutional/regulatory quality; community acceptance and participation; equity in energy access; renewable energy potential/ energy efficiency. The dimensions are not simply a collation of indicators found in isolated studies, but rather are the outcomes of thematic aggregation of the recurring patterns/constraints emerging from the literature. The contribution of previous literature then becomes more than just a direct list of criteria, but starts to be one that explains the structure and the relationships between the elements of the structure, intending to support the development of this structure.

The innovation of the research is that instead of focused analyses of parts of the energy transition, this one is done in an integrated way with adaptation. Yet another framework identifies what is currently in place (and therefore ready for implementation), what needs to be fundamentally changed, and what requires a higher level of preparedness. It combines ideas of Energiewende, the Multi-Level Perspective, and smart city technologies in a

phased pathway that connects with institutional and regulatory transition and infrastructure, distributed energy and demand management, and community involvement. Therefore, the German experience does not simply invite the adoption of a model, but rather offers principles and functions (reformulated as analytical variables) which have to be assessed in the Iraqi context for adoption.

2. Theoretical and Conceptual Framework

2.1 The Concept of Smart Cities: Definition, Components, and Challenges

A smart city is defined as a city that adopts information and communication technology and modern innovation to facilitate its inhabitants' way of life and benefits the performance of public services and infrastructure, aiming for sustainability across social, economic, environmental, and political aspects and understanding the specific needs of a city at any given time (Chatterjee et al., 2021). The focus is on how to use information, analytics, and digital technologies effectively for enhancing residents' experience, optimizing urban operations, and focusing on sustainability (Gracias et al., 2023). With not only the very rapid transformations in urban systems, demographic acceleration, and growing urban demand for energy, services, and resources, but also with the need to become more efficient in city management, smart cities have gained interest.

In this sense, urban intelligence is not only gathering data on urban conditions, but also the skill of analysing these data and embedding them into planning and decision-making processes, enabling much quicker responses to the changing urban needs (Batty, 2013). As such, the overall vision of a smart city is not just a city with technology, but rather one that integrates human and social capital, digital smart-technologies infrastructure, and institutional capability, and that is defined by the degree of co-ordination, participation and decision-making. Therefore, there is more intelligence in a city when these technologies help to coordinate, participate, and make informed decisions rather than in tracking the number of technologies deployed.

A smart city offers an opportunity to connect urban systems with each other, meaning that energy, transportation, buildings, and services are coordinated and managed in a smart manner with interconnected data exchange. The digital interaction tools also present further opportunities for the community and local institutions to exchange information, facilitating participation and quicker response, and increasing the city's resilience to shocks and stresses (Albino et al., 2015; Batty et al., 2012).

Smart cities can be considered as an enabling environment for energy transition in this study, as they offer the tools for real-time monitoring, demand management, integration of distributed energy sources, and coordination of energy, transportation, building, and governance systems. Digitalization and decentralization, however, do not necessarily result in transition success—their use requires a reliable network, high-quality institutions, financial capacity, cybersecurity, social acceptance, and equitable access to technologies.

This study can therefore not be interpreted as a blueprint of how Germany can be used as a model for Iraq. Rather, the German ideas are utilized as an analytical yardstick for analysis of components that can be utilized in Iraq and others that must be adopted and changed considerably.

Building on this insight, a smart city consists of a combination of elements, services, and strategic components interconnected with each other, which can be summarized as seven elements (Bellini et al., 2022; Trincă, 2024):

Technological infrastructure: This encompasses the deployment of Internet of Things (IoT) sensors, wireless networks, smart meters, and digital platforms, along with their management and oversight to monitor and control urban systems, while linking energy, transportation, and building services.

Data analysis: Involves gathering and interpreting data provided from different sources to monitor energy use, projection and forecasting, and to optimise urban performance and facilitate good decisions.

Digital communication and participation: This involves opening up for digital communication between institutions and the population, as well as engaging the community in planning and decision-making, capturing feedback and complaints, and granting transparency and trust.

Smart transportation: This involves optimizing the efficiency of public transportation; traffic management systems; electric mobility and charging infrastructure; transportation and renewable energy systems, etc., so as to minimize congestion, fuel consumption, and emissions.

Cybersecurity and data governance: This component should involve technical and institutional mechanisms for the prevention of data breaches, data misuse, safeguarding privacy, ensuring continuous operation of services, and enhancing trust in smart systems.

Environmental and energy sustainability: Energy efficiency, integration of renewable energies, emissions reduction and resource-, waste- and water management.

Urban innovation and experimentation: This means that pilot projects, start-ups, universities, and local projects involving new solutions in the field of distributed energy and storage, demand management, and smart services, are prepared.

In this study, these components are not treated as a predetermined set of technical characteristics, but as enabling capabilities whose relevance depends on the local context. In the Iraqi case, leveraging digital infrastructure, data

analytics, and distributed energy depends on reliable electricity and communications, institutional capacity for data management, financing, cybersecurity, social acceptance, and equitable access to digital services. Therefore, these components are subsequently used to assess the urban environment's readiness to support energy transition, not to measure the overall level of a city's intelligence. These relationships are illustrated in Figure 1.

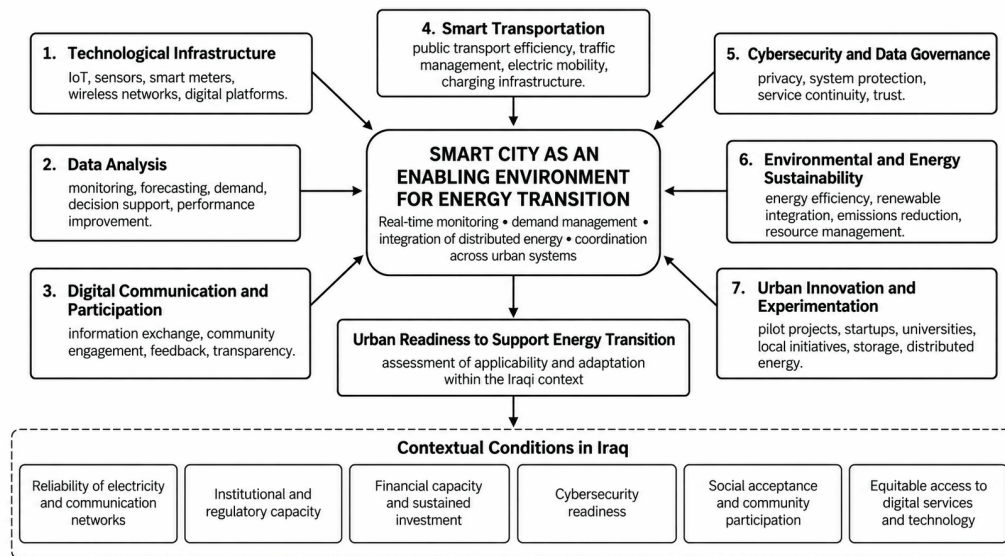


Figure 1. Integrated framework of smart city components as enabling capabilities for energy transition in Iraq

Urban areas face numerous challenges in their smartness initiatives, which can be summarized into two interrelated groups: establishment and adoption challenges, and challenges of operation and sustainability. During the establishment and adoption phase, social and cultural challenges emerge (Key social and cultural challenges include limited awareness of smart technologies, weak trust, disparities in digital skills, and insufficient public participation in planning and decision-making). Furthermore, there is the challenge of cybersecurity and protection against hacking, fraud, and misuse, and ensuring privacy, service continuity, and trust in smart systems. Some smart city applications may require adjustments to spatial planning and land use to provide digital infrastructure, charging stations, distributed energy facilities, and smart transportation networks, while avoiding imposing uniform spatial changes that do not take into account the local characteristics of the city. Finally, the implementation of smart cities also requires a review of planning, infrastructure, energy, and data laws to enable the integration of new technologies and to define the responsibilities of relevant stakeholders. At the same time, adoption is influenced by the extent to which residents accept and utilize digital services, particularly the elderly, those with low incomes, residents of underserved areas, and those with limited digital skills (Hollands, 2008).

The main concerns in the operational and sustainable phase are continuity of services, efficiency of systems, and the services and resources needed for long-term system modernization and sustainability. Smart systems sustainability is not just about the initial capital cost and relies on long-term operation and maintenance costs and future upgrading costs. Administrative flexibility and technical competences to address interrelated urban systems and aid in sharing data between various sectors are also needed. It involves good co-ordination between the disciplines, such as planning, energy, transport, communications and IT-related disciplines, and the ability to bring the models into practice, that is, conditions in the city and not a set of models. Therefore, a 'smart city' should not only be perceived as a technical one which requires the installation of infrastructural systems, but rather as an institutional process, or even institutional evolution with periodic updates, capacity building and evaluation of performance and learning from experience. It must, however, be continually assessed and evolved with active user engagement generating acceptance, trust and adaptability of the technology and associated institutional change and the promotion of the digital skills of the user base (employees and residents).

Handling smart-city development in this context means that it is a continual process of technological, institutional and social adaptation instead of a one-off project (Albdiery & Al-Mosawy, 2024a; Batty et al., 2012; Behzadfar et al., 2017). But in the context of Iraq, these challenges can be intensified due to unreliable electricity and telecom networks, the presence of many responsible authorities, poor coordination among institutions, the lack of long-term financial support, and differences between cities and social groups in terms of digital smarts. Therefore, these challenges are treated as critical conditions that determine the viability, scalability, and sustainability of smart city technologies in supporting the energy transition. Figure 2 presents an analytical classification of smart city challenges into establishment and adoption challenges and operational and

sustainability challenges, with an explanation of the factors that exacerbate them in the Iraqi context.

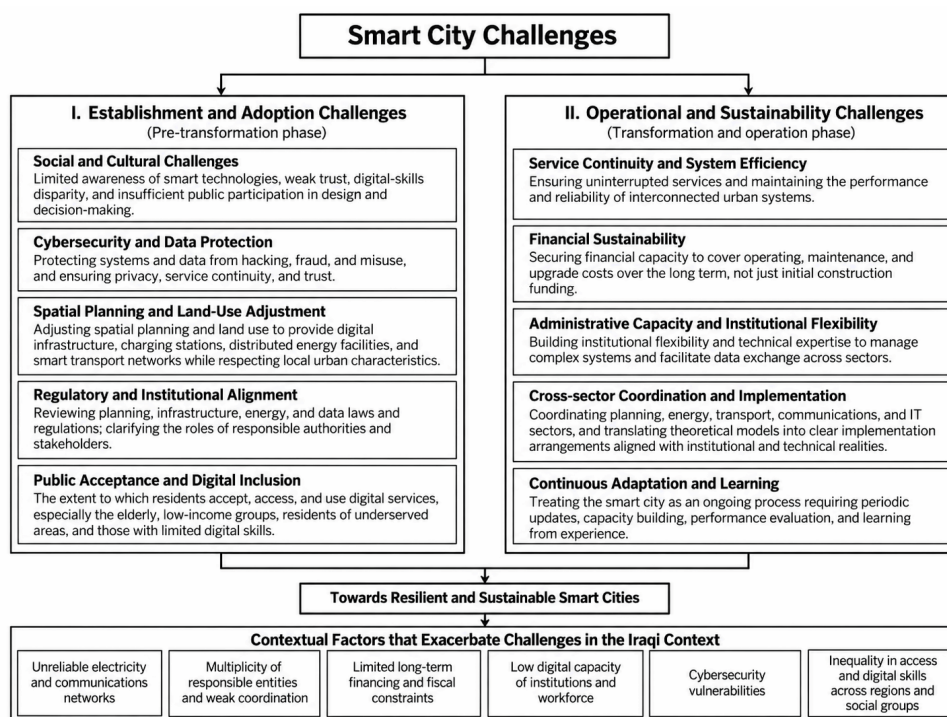


Figure 2. Analytical classification of smart city challenges into establishment and adoption challenges and operational and sustainability challenges

Source: Prepared by the authors based on Batty et al. (2012); Behzadfar et al. (2017); Hollands (2008).

Efforts to reduce urban carbon emissions have led to the emergence of ambitious projects and strategies for developing low-carbon cities. Many low-carbon city models are associated with relatively compact urban forms, shorter commute distances, and greater walkability and cycling accessibility, along with the integration of land uses with public transport and energy efficiency in buildings and infrastructure (Ye et al., 2018). Focusing on compact neighborhoods does not mean shrinking the city, but rather improving connectivity with efficient and multimodal public transport networks, which can reduce the need for long journeys and excessive reliance on private vehicles (Yahia et al., 2026). These models aim to reduce energy demand, improve energy efficiency, and gradually decrease reliance on fossil fuels by expanding renewable energy sources, developing low-emission transport systems, and improving the energy performance of buildings (Esfandi et al., 2024).

The integration of renewable-energy systems and sustainable transport, together with improvements in energy efficiency and spatial planning, represents a key component of urban greenhouse-gas mitigation strategies (Shayan & Ghasemzadeh, 2020). In this study, the principles of low-carbon cities are not treated as a unified spatial model directly transferable to Iraq, but rather as a set of planning principles that include reducing energy demand, improving building efficiency, integrating land use and transportation, and expanding renewable energy. These principles will largely depend on the current urban fabric and the condition of the public transport system, as well as grid availability and institutional and financial capabilities to coordinate energy, transport and urban planning.

A prominent feature of national strategies for developing smart and socially responsible cities is the growing emphasis on renewable energy and sustainable or low-emission mobility, particularly when energy, transport, and urban-planning policies are integrated within a coordinated framework (Bibri, 2020). The development of electric mobility relies on building a supporting infrastructure—including charging points, electricity grid reinforcement, and smart demand management—and more renewable sources of electricity to power charging stations (Statharas et al., 2019).

However, if accompanied by efficient public transport, a less fossil fuel-intensive electricity mix, and policy facilitating the anticipated uptake and acceptance of these developments, these can help to boost energy efficiency and reduce emissions, and foster more sustainable patterns of mobility (Silva Cruz & Katz-Gerro, 2016).

The promotion of electric vehicles in the Iraqi context should not be treated as a stand-alone priority, but rather its usefulness hinges on the reliability of the electric grid, generation capacity, distribution of charging stations, the cost of vehicles, and equal accessibility. Hence, the proposed approach considers electric mobility as something that can be scaled up in parallel to grid modernization, as well as the growing share of renewables, instead of something that can be realized at a large scale in the first phase.

Hybrid solar-energy storage systems combine solar generation with energy-storage technologies to store excess electricity for use during periods of low generation or high demand (Buonomano et al., 2018). Electricity generated from solar energy can be stored in batteries for short- or medium-term use or converted to hydrogen through electrolysis for longer-term storage in selected applications (Hasan et al., 2023). An integrated photovoltaic (PV) and storage—whether battery or hydrogen—can mitigate the impact of the intermittency of PV generation, make the PV system more flexible, and add to the reliability of the electricity system. An appropriately designed and operated hybrid solution can offer different storage durations, increase EV charging flexibility, alleviate EV battery stress, and reduce the dependence on diesel generators and resulting greenhouse gas emissions (Coppitters et al., 2020). The importance of storage systems, particularly in the Iraqi context, is related to the variation in the power supply from electricity, high reliance on the usage of diesel generators, in addition to the high solar potential. But its growth and penetration depend on the cost, availability of maintenance and technical expertise, and adequate capacity of the grid to support distributed systems and some sort of regulatory structure that supports interconnections and electricity sharing. Accordingly, the proposed framework treats storage as a transitional solution that can initially be applied through pilot projects and microgrids and expanded gradually as institutional, financial, and infrastructural readiness improves.

2.2 Definition of Photovoltaic Systems

A PV electricity-generation system consists of semiconductor cells assembled into panels that convert solar radiation into direct-current (DC) electricity, which is then converted into alternating-current (AC) electricity by an inverter (Jieb & Hossain, 2022). These systems can be used as stand-alone or grid-connected, and they yield renewable electricity with low operating emissions and with a relatively small amount of maintenance compared to a conventional electricity plant. The growing digitization of urban services, smart buildings, electric mobility, and data-driven digital infrastructures in smart cities is creating an increasing demand for reliable, low-emission electricity supplies (Mao & Ni, 2024). The further reliance on power production based on fossil fuels is seen as economically and ecologically unsustainable. PV technologies can support the decentralization and decarbonization of electricity generation—either in building-scale, neighborhood, or microgrid applications. PV can be used in many scales: large-scale solar power plants or distributed systems placed on a building's roof or built into an urban infrastructure element, depending on the available space, size of the grid, and economic viability (Ayaz et al., 2020).

PV systems can be deployed at both centralized and distributed scales, while Iraq's high solar radiation provides substantial potential for integration into the energy transition. Here, the importance of transmission and distribution network upgrading, regulating grid connections, creating tariff and purchasing mechanisms, providing financing and maintenance options, and also including storage and regulating demand emerge as keys to scalability. Accordingly, PV will be treated as an early-start element; its rollout will be incremental, followed by distributed PV to larger PV applications as the institutions, network, and financial readiness increase.

2.3 Energy Transition “Energiewende”

Energiewende is a German term referring to a comprehensive transformation of the energy system toward greater sustainability through changes in energy sources, energy use, regulation, and system organization. This transformation involves the gradual reduction of fossil-fuel dependence, expansion of renewable energy, improved energy efficiency, network modernization, decentralized generation, and demand management. In the German context, it also includes the phase-out of nuclear power (Kalair et al., 2021; Scholz et al., 2014).

Energiewende in this study is not a specific set of time-constrained targets or a prescribed course of action but a collection of institutional, technical, and social principles which can be tested for their adaptability to Iraq. Thereby, a difference is created between general, applicable principles, the prerequisite conditions for the use of tools that have to be made more suitable to the Iraqi context, and goals that are specific to the German historical and institutional context. Under the Energy Transition, considerable updates have been made to the legal and policy goals related to GHG emission reduction, climate neutrality, the share of energy from renewable sources, and energy efficiency in Germany's Electricity Sector. Table 1 summarizes the revised key goals of Germany's energy transition, taking into account the existing German law (Federal Republic of Germany, 2019; Federal Republic of Germany, 2023a; Federal Republic of Germany, 2023b). The table not only specifies a set of timelines for each of these objectives but also describes each objective, its relevance, and applicability to the context of Iraq. As shown in Table 1, the Energiewende cannot be transferred directly to Iraq by focusing on the same legal targets or timeline, but rather its goals, principles, and the structure of the energy system must be interpreted to benefit the Iraqi context, with relevant configuration to its institutional, financial, and social capacities. Given that, the study puts forward a sequential framework for the overall target of reforming the Iraqi energy system, starting with more feasible early-stage priorities at the present stage, and moving forward to the less achievable ones that include increasing the reliability of the grid, investment in renewable energy capacity, building institutional capacity, and

securing public approval. These phases are not defined by fixed timeframes or numerical thresholds; rather, they represent an adaptive sequence in which progression depends on improvements in Iraq's technical, institutional, financial, and social readiness.

Table 1. Legal objectives of the Energiewende and their implications for the Iraqi context

No.	Dimension of Energy Transformation	German Legal Objective	Its Nature and Legal Source	Adaptability to the Iraqi Context	Analytical Implications for the Iraqi Context
1	Reducing greenhouse gas emissions	Reduce emissions by at least 65% by 2030, and by at least 88% by 2040, compared to 1990 levels	Binding legal objective; German Federal Climate Protection Act, Section 3(1)	Adaptable in principle, but not in terms of proportions and schedules	Iraq should implement a step-wise reduction of emissions, but national baselines and phased targets should be set based on the level of data reliability, economic structure, and capability of the different sectors.
2	Climate neutrality	Achieving net-neutral greenhouse gas emissions by 2045	Binding legal objective; German Federal Climate Protection Act, Section 3(2)	It needs fundamental modification	It can be adopted as a long-term strategic direction, not as a German deadline that can be transferred directly to Iraq.
3	Electricity produced from renewable energy	Increase the share of electricity produced from renewable sources to at least 80% of total electricity consumption by 2030	Legal objective: Renewable Energy Sources Act EEG 2023, Section 1(2)	The principle is adaptable, but the target share and deadline are not directly transferable	Iraqi applications involve the development of renewable energy in parallel with addressing the supply deficit, reduction of losses, upgrading of the grid, and also storage and demand management.
4	Ultimate energy efficiency	Reduce ultimate energy consumption by at least 26.5% by 2030 compared to 2008	Legal objective: German Energy Efficiency Act, Section 4(1), no. 1	Highly adaptable	It is possible to start early with standards for the efficiency of buildings, equipment, lighting, and public services that are not transferred from Germany, but that use an Iraqi baseline.
5	Primary energy efficiency	Reduce primary energy consumption by at least 39.3% by 2030 compared to 2008	Legal objective: German Energy Efficiency Act, Section 4(1), no. 2	Adaptable after adjusting the indicators	Iraq needs indicators that are consistent with its generation structure, network losses, fossil fuel consumption, and power plant efficiency.
6	Long-term energy efficiency	The goal is to reduce final energy consumption by 45% by 2045 compared to 2008	Long-term approach; German Energy Efficiency Act, Section 4(2)	This is a guiding principle, not a directly transferable goal	It can be used to guide the long-term course, provided that the ratio and duration are determined according to Iraqi economic and institutional capabilities.
7	Phasing out coal	Reduce the remaining net capacity of hard-coal and lignite-fired power plants to zero gigawatts no later than the end of 2038	Legislative objective: Coal-Fired Power Generation Termination Act, Section 4(1)	Limited relevance to the Iraqi context	Coal is not a major component of Iraq's electricity mix; however, the general principle of phase-out of higher-emission sources can be utilized.
8	Phase-out of nuclear power	Germany completed the phase-out of nuclear power generation as part of its national policy trajectory	An element specific to the German experience, not an existing future goal	Non-transferable	It is linked to German political and technological history, and does not represent a direct priority within the Iraqi energy transition.

3. Methods and Materials

3.1 Research Design and Analytical Approach

Accordingly, a comparative descriptive-analytical design using secondary data analysis was used for evaluating the suitability/adaptability of the main Energiewende principles/policy instruments in the Iraqi context. The Energiewende experience was used as an analytical reference case, while the Iraqi energy system was used as a contextual assessment case, where the demands, constraints, and context of its application were studied concerning the application of the Energiewende principles.

The principles, institutional arrangements, and technical and social instruments associated with Energiewende were taken as units of analysis, including renewable energy, energy efficiency, network modernization, decentralization of energy generation, demand management, regulatory frameworks, and community involvement.

Many of these were adaptable to Iraq, and it was the characteristics of the Iraqi energy system, as well as the institutional, financial, technical and social barriers that they faced, that allowed us to evaluate the adaptability of these elements in the absence of direct transfer or identical application of the German model.

Analysis was restricted to a mutually coherent set of readiness conditions and contextual factors, including electricity grid reliability, investment capacity, institutional and regulatory quality, community acceptance and participation in the system, equity of access to energy services, potential for renewables, energy efficiencies and system accommodation for distributed generation, storage and demand management.

3.2 Study Context and Scope: The Iraqi Energy System

This study analyses how the electricity system functions in Iraq (in terms of generation, transmission and distribution) and the institutional, regulatory, financial and social context in which the electricity transition can take place. The electricity system in Iraq has been undermined by decades of underinvestment, conflict and unstable conditions and has come under strain as the population grows and the nation expands significantly. Rising demand and rapid urbanisation have added strain to Iraqi generation, transmission and distribution networks, which have been undermined by decades of underinvestment, conflict and unstable conditions. All these have contributed to burdening generation, transmission and distribution networks with frequent outages and increased demand for private generators, as well as limited efforts to leverage the adoption of rooftop solar energy systems (Albdieri & Al-Mosawy, 2024a).

The use of older fossil-fuel technologies, such as diesel- and gasoline-powered processes, sulfur dioxide injection and additive treatment, and steam generation, etc., is significant and remains one of the main sources of carbon emissions in Iraq due to its reliance on fossil fuels and sparse development of renewable resources (Al-Mosawy et al., 2021; Al-Rikabi et al., 2026; Bekheet et al., 2023). However, on the other hand, there is a high potential for solar energy, with Iraq's high solar irradiance values and long annual sunshine hours, offering a favorable natural background for enhancing solar energy power generation and distributed solar renewable-energy projects (Abed et al., 2014).

However, uptake of renewable-energy technologies has been slow given Iraq's massive natural potential and increasing difficulties in the energy sector (Ersoy & Terrapon-Pfaff, 2021). High natural potential in Iraq coupled with technical, institutional, financial, and social constraints makes Iraq an appropriate case for assessing the adaptability of Energiewende principles (Hassan et al., 2024).

Germany was not treated as a direct template for Iraq, because the two national contexts are not directly comparable. Instead, the Energiewende experience was used as an analytical reference to identify relevant principles, instruments, mechanisms, and implementation requirements. For this reason, functions and principles that can be analytically separated from the specific German context were selected from the specifics of the German context by considering differences between Germany and Iraq in network structure, institutional and financial capacities, market regulation, public and stakeholder acceptance, and energy security priorities.

The scope of the study was largely restricted to the electricity sector and its links with renewable energy, energy efficiency, smart grids, distributed generation, and demand management. No attempt was made to evaluate any particular development project or city in Iraq or even all the energy sectors in detail. Rather, it was the question of the national requirements for applying energy transition principles and the enabling role of smart urban planning that was explored.

3.3 Data Sources and Document Selection

Secondary data from four main categories of sources were used in the study. German official legislation, policies, and strategies on energy transition, emissions reduction, renewable energy, energy efficiency, and electricity market regulation were included in the first category. The second type of records included Iraqi government documents, strategies, and reports regarding the electricity sector structure, electricity generation and demand,

reliability of the electricity grid, the efforts of the Iraqi government in implementing renewable energy policy, and its institutional and financial challenges. The third category contained reports from international organizations related to energy, development, and sustainability, and the fourth category contained peer-reviewed scientific studies that focused on the Energiewende experience, the transfer and contextual adaptation of energy policies, decentralized generation, smart grids, public and stakeholder acceptance, and equitable access to energy.

Selection criteria required that sources originate from official bodies, recognized international organizations, or peer-reviewed scientific journals and provide data or analysis relevant to one or more of the study dimensions. These included having a source from an official body, a recognized international organization, or a peer-reviewed scientific journal, and that it contained data or analysis in one or more of the dimensions studied and information that could be used to compare with or make a contextual assessment of the Iraqi energy system. In cases of energy transition or the Multi-Level Perspective, more recent sources were given more weight when analyzing current goals and policies, and some older studies were included when they provided a foundational reference for understanding the evolution of energy transition or the Multi-Level Perspective.

Documentation that did not provide in-depth information on the energy transition but rather general information, without being related directly to the selected indicators, was excluded. German sources were not directly used to prove applicability to Iraq, but as a source to distil out principles and requirements that were then broken down and discussed within its context.

The aim of source collection was not to provide an in-depth literature review but instead to provide a systematic evidence base to facilitate extraction of Energiewende elements, readiness conditions and Iraqi constraints, and comparison, classification and development of a phased framework.

3.4 Analytical Dimensions, Indicators, and Evaluation and Classification Criteria

The analytical dimensions have been identified based on the German energy transition experience, starting with the fundamental principles and tools, and their association with the research questions and the unit of analysis. The sensitivity levels were then compared with the constraints and level of readiness, based on Iraqi documents, international reports, and relevant scientific studies. The “energy transition” in Germany (the principle, policies and instruments) was not considered a “turnkey” solution that can be directly applied to Iraq. These were, rather, decomposed into key functions and technical, network, financial, institutional, regulatory, and social needs. These were then evaluated based on the functions/needs and their adaptability to the Iraqi system.

This process resulted in identifying a set of overlapping elements and grouping them into six analytical dimensions: technical and network readiness, financial and investment capacity, institutional and regulatory quality, community acceptance and participation, equity in access to energy, and renewable energy potential and energy efficiency. As for the elements related to the specificity of the German experience, such as phasing out nuclear energy and the legally defined timetables for the transition, they were not treated as transferable indicators, but were used as contextual references that help to define the limits of comparison and to differentiate between adaptable basic functions and elements related to the German historical, legal and institutional context.

Each dimension was operationalized through a set of qualitative and quantitative indicators, which were then used to assess each energy-transition element across the six dimensions to determine the level of need, the scale of available opportunities, or degree of constraint associated with implementing each element of the energy transition. Quantitative indicators, based on data availability, included the level of fossil fuel dependency, electricity demand growth, the supply-demand gap, transmission and distribution losses, the contribution of renewable energy, and solar irradiance potential. Qualitative indicators were the stability of policy; clarity of responsibilities for interconnection licensing and tariff procedures; finance, institutional capacity for implementation and monitoring, community acceptance and participation, and equity in access to energy services and relevant technologies. The intensity of the need or size of the opportunity was used to identify through quantitative indicators, and the conditions of implementation and institutional, financial, and social constraints that may impact the feasibility and scalability were identified through qualitative indicators.

They were not put into a composite index or an overall numerical score, nor were they divided by numerical weights or statistical tests, as that was not the purpose of the study—rather, to assess and recognize the role of each. Rather, they are presented as a set of structured qualitative indicators to gauge whether or not the element is suitable for direct transfer, the readiness for implementation, and the technical, institutional, regulatory and/or financial changes that would be necessary, and the risks involved in its direct transfer. Based on this assessment, elements of the energy transition were split into three categories: highly adaptable elements, elements requiring substantial modification, and elements specific to the German context or dependent on the conditions of an advanced energy system which are not directly an energy transition priority for Iraq at its current stage. These elements were also linked to the early, intermediate, and advanced stages of the proposed transition path according to the level of readiness and critical constraints, rather than to fixed timeframes or predetermined numerical limits, as illustrated in Table 2. While Appendix Table A1 shows the application of these dimensions to each element of the energy transition, Appendix Table A2 shows the quantitative and contextual evidence, critical constraints, and

justifications on which the final classification of each element was based.

Table 2. Analytical dimensions and operational indicators used in assessing the adaptability of Energiewende elements to the Iraqi context

No.	Analytical Dimension	Main Indicators	Nature of Evidence and Its Use	Function in the Assessment
1	Technical and network readiness	Grid reliability, transmission and distribution capacity, storage, demand management, and capacity to accommodate distributed generation.	Quantitative data relating to production, demand, losses, and network capacities are used to determine the severity of the need or degree of the constraint, and are supplemented by documentary evidence relating to the state of the infrastructure, technical standards, and operational capabilities.	Determining the extent of technical applicability and scalability
2	Financial and investment capacity	Financing, feasibility, incentives, institutional investment capacity, and consumer affordability. Policy stability, clear responsibilities, regulation of grid connection and tariffs, institutional coordination, and the capacity for implementation and monitoring.	Data relating to costs, investment, and financing are used to determine the size of the financial gap, and are complemented by an analysis of incentives, risks, safeguards, and affordability.	Assessing feasibility and financial sustainability
3	Institutional and regulatory quality	Public and stakeholder acceptance, trust, participation, and willingness to adopt technologies.	The evidence relies on legislation, policies, institutional reports, and studies that demonstrate the clarity of jurisdictions and procedures, as well as the capacity for implementation and enforcement.	Identifying the need for institutional and regulatory reform
4	Community acceptance and participation	Affordability, geographic distribution, energy poverty, and access to energy-related digital technologies and services.	Evidence drawn from studies and reports on awareness, trust, participation, and experience in using decentralized solutions is used, taking into account the limitations of direct field data.	Assessing social and political sustainability
5	Equity in energy access	Availability and potential of renewable energy resources, building and transport efficiency, energy loss reduction, and consumption management.	Data relating to service disparities, financial burdens, and access to energy are used, supplemented by contextual evidence relating to low-income groups, underserved areas, and the digital divide.	Assessing the fairness of the transition and ensuring that vulnerable groups are not excluded
6	Renewable energy potential and energy efficiency		Quantitative data relating to solar radiation, renewable energy contribution, demand, and losses are used to determine the size of the potential, and are read in conjunction with the technical, financial, and institutional constraints that determine investment feasibility.	Identifying opportunities for early intervention and reducing demand and emissions

To ensure consistency in the assessment and to allow for tracing the basis upon which each element's classification was based, a qualitative analytical matrix was used to evaluate the energy transformation elements across the six dimensions. The impact of each dimension on the applicability of the element was assessed using three ordinal states: supportive, conditionally supportive, and constrained. Elements related to the specificities of the German context or those not of direct Iraqi priority were addressed within the final classification of elements, not as independent assessment cases within the six dimensions.

A "supportive" assessment indicates that current or near-current conditions allow for the initiation of the element's implementation, and that the required modifications are limited or manageable. A "conditionally

supportive” indicates that the element can be implemented on a pilot and limited basis, but requires certain capability enhancements in the network, funding, regulation, or institutions. “Constrained” means there is a significant lack of the respective dimension and suggests that the widespread application is not possible or that a direct transfer is not suitable.

This final classification was done in a logical manner according to critical constraints, and not according to a simple average of the dimension assessments. This logic finds that if an element has a technical, institutional, or regulatory barrier that hinders implementation of this element even if the others are supportive, then it is not considered highly adaptable. If an element has a direct Iraqi need; most dimensions have a “supportive” or “conditionally supportive” function; and no dimension is a critical constraint preventing initial application; then that element is deemed to be highly adaptable. Elements whose basic function is appropriate for the Iraqi situation but whose implementation needs significant changes in the network, funding and/or institutional or regulatory framework are placed in the category of elements that require substantial modification. When elements do not directly relate to the priority level for Iraq that is used in the present stage, they are included in Category 3 when they do address details of the German experience, or requirements of an advanced energy system.

3.5 Comparative and Contextual Adaptation Procedure

The comparative analysis has been done at two interrelated levels. The first step was to identify the basic functions and requirements of each *Energiewende* principle and/or the policy instruments. This included their objectives, institutional and regulatory arrangements, and technical, financial and social requirements. The second level explored the degree to which these conditions exist in the Iraqi context, identifying convergence, gaps, and constraints for applicability.

The qualitative dimensions and indicators from Table 2 were used to assess the elements of *Energiewende*. Fit of each element with the needs of the Iraqi energy system and technical and network readiness, financial and investment capacity, institutional and regulatory frameworks, public and stakeholder acceptance, as well as potential implications for equitable access to energy were taken into consideration. The energy potential of renewable energy resources, the potential of energy efficiency, and the threat of implementation without adaptation in the context were also considered.

The evaluation of adaptability was oriented towards the core function of the element and the institutional or legal form through which its implementation in Germany took place. Elements whose basic use fits Iraq’s needs and can be introduced with limited modifications were considered highly adaptable. Those that needed significant changes to the network, finances, institutional arrangements, governance, or community participation were considered to require substantial modification. Objectives, decisions and timelines that referred exclusively to the German experience and were not a direct priority for Iraq were considered elements specific to the German context.

This classification was not considered a final assessment, but rather a contextual assessment based on the present level of readiness. These elements were then associated with suggested stages of the energy transition. Those elements requiring relatively limited institutional and technical resources were given priority for the early phases, and those that require a more stable network, more investment, and more advanced institutional and regulatory capacities were given preference for the intermediate or advanced phases. The assessment did not use fixed quantitative weights or numerical thresholds; rather, a comparative interpretation of the documentary evidence was used. Hence, no conclusive statistical measure of the results is presented, but rather contextualized analytical classifications.

3.6 Integration of the Multi-Level Perspective

The Multi-Level Perspective provided a structured interpretive framework to analyze the energy transition as the interaction of external pressures, the existing system structure, and emerging innovations. It is not applied as a separate theoretical addition but is integrated into the comparative and classification processes to illustrate the differing degree of adaptability of the elements and the conditions required for their inclusion in the proposed transition stages.

Three interrelated levels were created for the documentary evidence and analytical observations. The socio-technical landscape level represents the long-term external pressures impacting the Iraqi energy system, including reliance on fossil fuels, growing electricity demand, climatic and economic pressures, and energy security requirements. The socio-technical regime level comprised established institutions, rules, and practices such as the grid structure, subsidy and tariff mechanisms, the distribution of responsibilities, and regulatory and financial capacities. The niche-innovation level included technological and social alternatives, notably distributed solar power, microgrids, storage, demand management, smart meters and local initiatives—technologies and social arrangements which could be piloted and then scaled up.

This classification was employed to provide an explanation as to where each element is in the transition process. Areas that might be considered small-scale or pilot innovations, or could be implemented on a smaller scale in

pilots, were identified as emerging innovations appropriate for the early stage. Those elements that depended upon tariff reform, network modernization, the redistribution of institutional roles, or the development of market rules were linked to systemic change and integrated into intermediate and advanced stages. The pressures from the wider landscape were also invoked to explain the factors behind the energy transition and conditions that may facilitate or constrain the energy reform process.

No set thresholds were set between the levels because some innovations might transition from the emerging to the established system as they developed and had institutional and regulatory frameworks. Therefore, use of the Multi-Level Perspective was beneficial in linking the categorization of context and the proposed transition phases and to differentiate between interventions at the pilot level and structural changes and outside pressures to a transition.

3.7. Development of the Phased Adaptive Framework

The phasing and adaptive framework resulted from the synthesis of the comparative assessment results, the adaptability classification, the readiness-condition analysis, and multi-level interpretive process. It wasn't designed to mimic the transition approach of Germany, but rather to offer an approach that considers the adaptive principles and requirements of the Iraqi energy system.

All elements were added to the framework based on their adaptability, readiness for intervention, needed reforms and resources, and their dependence on previous interventions as well as relevance to current challenges in Iraq.

Early interventions were done that began to tackle the problems within existing constraints and did not require genuine "subversion" of the existing socio-technical regime. In the intermediate phase, interventions would have to relate to enhanced grid reliability, investment capacity, and institutional capacity as well as the establishment of regulatory frameworks. All required a greater level of digital integration, community participation, institutional and regulatory capacities in the advanced phase, as well as greater grid stability and long-term financing.

While the phases were not designed as fixed time periods or as quantitative thresholds, they were crafted as an adaptive process, with progression to the next phases dependent on improving the readiness conditions and on the readiness criteria being met, which include: grid reliability, investment capacity, institutional and regulatory quality, community acceptance, and equitable access to energy services. Some interventions may be in the form of an early pilot project, and then be scaled up once these conditions become better.

This outcome is an overall analytical model (not an actual national model or one which already has been tested or field validated) to prioritize and delineate implementation conditions. Thus, both its effectiveness and the suggested sequence of its components need to be validated in the future via interviews, questionnaires, quantitative analysis, scenario simulations, and pilot applications in the Iraqi context.

4. Results and Discussion

4.1 Contextual Assessment of the Iraqi Energy System

The contextual assessment revealed substantial potential to develop Iraq's energy system towards a more sustainable way of transition, but also identified certain structural barriers that make the transference of the German energy transition model difficult. The key strengths are distributed generation and the high potential of solar energy; the key weaknesses are a low share of renewables, high grid losses, low reliability of the current grid, still reliant on fossil fuels, and inadequate supply to meet increasing demand. The quantitative information collected, which underpins this, shows that Iraq had around 135–150 TWh of Electricity production annually, less than 5% of which was from renewables and 90%–95% from fossil fuels. Additional data of about a 5%–7% annual increase in electricity demand was also reported, and almost 70% share of the population. The quantitative facts given in Table 3 corroborate the dependency on fossil fuel-based power generation, the ineffectiveness of renewable generation, and the increasing strain from electricity demand. All these conditions suggest that system rehabilitation, loss reduction, and enhanced monitoring of the system should be done before going upon any extensive integration of renewable energy sources. But there are still numerous obstacles to be overcome in order to integrate distributed generation at a more massive level, including inadequate grid connectivity, lack of technical facilities for bidirectional electricity transmission, and insufficient protection and control systems (Al-Rikabi et al., 2026).

The big problem is not a lack of renewable resources; it's the disconnect between the potential for renewable energy in Iraq and how much renewable energy can be used by the current system. The electricity demand is also growing at the rate of 5%–7% per year, which further strains the already limited Public Electricity System. A more stable grid, demand management and storage, as well as metering and data capabilities, and institutional and regulatory reforms to facilitate a distributed generation grid are required as renewables are introduced in larger percentages. The reforms also include in their mandate the regulation of electricity producers and operators of the

grid.

Table 3. Key quantitative indicators of the Iraqi energy system

Indicator	Value	Unit	Interpretation
Total electricity generation	135–150	TWh/year	The volume of electricity production in Iraq
Fossil fuel share in electricity generation	≈90–95	%	Heavy reliance on oil and gas
Renewable energy share	<5	%	Limited use of renewable energy
Electricity demand growth	5–7	% per year	Indicates increasing demand due to population and urban growth
Urban population	≈70	% of total population	Reflects urban pressure on energy infrastructure
CO ₂ emissions from the energy sector	≈200	million tons/year	An environmental indicator showing the impact of the traditional energy system

The result also indicated that the dominant use of private generators can also be attributed to poor public electricity supply. The annual electricity-demand growth of about 5%–7% increases the gap between supply and demand, resulting in the existing dependence on private generators. Meanwhile, this reliance marks the need for decentralized energy solutions from a socio-economic standpoint (Altai et al., 2022). As long as the necessary regulating, financing, monitoring, and enforcement systems are in place, this current trend of decentralized electric power supply presents an opportunity to shift some of the local electric power supply to storage, distributed solar energy, and microgrids. Restrictions go beyond institutional and monetary ones; they also involve technical ones. This again confirms the results presented in this paper so far that the potential of renewables in the country is enormous but nevertheless has not been enough to cast aside institutional, regulatory, and investment barriers. Yet, inadequate structures around the rights of small producers, decentralized energy generation, and integrating such decentralized sources with distribution grids can limit widespread investment in rooftop solar energy and other forms of local energy production (Hassan et al., 2024). A need for institutional coordination, poorly established regulations for connectivity to the grid, tariffs and incentives, and lack of investment capacity, which make it difficult to translate renewable energy potential into feasible projects to be scaled up, were noted. An unadopted, direct transfer of Energiewende instruments instead of the prepared framework, however, could lead to goals and/or structures that are not yet at Iraq's readiness level.

The future outlook on energy transition in Iraq thus looks good: the plan is to add some 10–12 GW of solar power by 2030, which reflects energy transition policy commitment in Iraq. But its realization depends on the modernization of the electrical grid coupled with the mobilization of investments, enforcement of regulatory policies, and institutional coordination (Al-Kayiem & Mohammad, 2019). Based on the results, it is concluded that the best path forward for Iraq is to start with implementing energy efficiency, energy loss reduction, distributed solar power applications, pilot projects on how to implement energy storage and microgrids, and institutional and regulatory capacity building.

4.2 Classification of Energiewende Elements According to Their Adaptability to the Iraqi Context

A comparative analysis revealed that there are differences in the adaptability of the elements of Energiewende to the Iraqi context. Some can be started with the existing or very near future situation, some need extensive technical, institutional, or financial changes, and a third are related to the German situation or to the needs of an advanced energy system. This variation captures the diversity of the elements' abilities to respond to the needs of Iraq, the readiness required for each to be implemented, and the extent of any critical constraints associated with each element, in line with the rules used to assess the elements shown in Section 3.4.

4.2.1 Highly adaptable elements

This will encompass storage and microgrid pilot projects, energy efficiency, distributed energy, capacity building, metering and data systems, and distributed solar energy, along with reducing transmission and distribution losses and demand management. These elements feature both technical and economic viability, as they are directly the elements of imbalance in the Iraqi system, which necessitates expansion that must be based on generating potential and financing, technical capability, and regulation.

4.2.2 Requires substantial modification

This encompasses incentive tariffs, financial and market incentive measures, decentralized energy ownership in the community, big decentralized generation, digital sharing, and advanced integration of energy and smart urban planning. The basic services for which these elements are intended suit the requirements of Iraq, but their extensive rollout will necessitate groundbreaking changes in the grid, financial plans, and institutional and regulatory

processes.

4.2.3 German context-specific elements

This involves a wide electricity sector liberalization, flexible markets and advanced electricity trading; quick decarbonization with strict quantitative goals; and rapid conventional electricity generators phase-out. Particularly in the Iraqi context, such elements are not on the immediate radar as they rely on a stable grid, a developed market, regulatory and financial skills, or are otherwise tied to particular targets and timeframes in the German context.

The classification is not strict, and some elements can be shifted from one classification to the other due to the improvement in technical, institutional and financial readiness. The result of the final classification is shown in Table 4, and the detailed basis for the final classification is provided in the appendices.

Table 4. Classification of German energy transition elements according to their adaptability to the Iraqi context

Energy Transition Element	Primary Function	The Current Situation or Need in Iraq	Adaptability Category	The Required Modification or Condition	The Most Suitable Stage
Improving energy efficiency	Reducing unnecessary consumption and lowering pressure on generation and the grid	High demand and poor utilization efficiency in buildings, facilities, and some production activities	Highly adaptable	Setting efficiency standards, auditing energy consumption, and providing incentives for efficient equipment and buildings	Early
Reducing transmission and distribution losses	Increase the amount of available electricity without adding a similar generating capacity	High technical and non-technical losses and obsolescence of parts of the network	Highly adaptable	Network upgrades, improved maintenance, and enhanced measurement, collection, and monitoring capabilities	Early
Distributed solar energy	Diversifying supply sources, reducing pressure on the grid, and supporting local generation	High solar potential and limited, unregulated deployment of solar systems	Highly adaptable	Establishing technical standards for installation and connection, financing mechanisms, and regulating the relationship with the network	Early then middle
Demand Management	Reducing peak loads and improving the balance between supply and demand	A gap exists between supply and demand, and there is weak control over consumption patterns	Highly adaptable	Developing tariffs, data, awareness programs, and demand response gradually	Early then middle
Smart metering and data systems	Improved monitoring of consumption, billing, planning, and network management	Limited accurate data and weak measurement systems in some areas	Highly adaptable	Starting with pilot areas, developing digital infrastructure, protecting data, and training staff	Early
Building institutional and technical capacities	Improving planning, implementation, follow-up, and coordination between entities	Disparities in capabilities, weak institutional coordination, and a lack of some specialized expertise	Highly adaptable	Training programs, defining responsibilities, enhancing coordination, and follow-up mechanisms	Early and continuous

Pilot storage projects	Increase system flexibility and support the integration of variable renewable energy	Limited application of storage technologies and relatively high costs	Highly adaptable within an experimental range	Choose clear locations and use cases, and assess feasibility and performance before expanding	Early experimentally, then intermediate
Microgrids	Improving supply reliability in specific areas or facilities	The need for local solutions in areas with poor supply or critical loads	Highly adaptable within a defined range	Setting standards for operation, connectivity, and ownership, and selecting suitable pilot sites	Early experimentally, then intermediate
Incentive tariff for renewable energy generation	Encouraging households, institutions, and investors to produce renewable electricity	Absence or incompleteness of tariff, procurement, and linking mechanisms	It requires a fundamental change	Designing a tariff that is compatible with financial capacity, and ensuring reliable measurement and a stable payment mechanism	Intermediate
Financial and market incentives	Reducing investment risks and accelerating the adoption of clean technologies	Limited funding, weak incentives, and an unstable investment environment	It requires a fundamental change	Directing incentives towards priority projects, avoiding unsustainable financial burdens, and improving transparency	Intermediate
Community or cooperative ownership of energy	Expanding local participation, sharing benefits, and promoting acceptance	The limited legal frameworks, institutional expertise, and funding available for this type of project	It requires a fundamental change	Developing simplified legal and financial models, and implementing local pilot projects	Intermediate then advanced
Large-scale decentralized generation	Increase product diversity and improve system flexibility	The presence of unregulated private generation, versus the network's limited capacity to accommodate multiple flows	It requires a fundamental change	Organizing connectivity, developing protection and control, and defining the responsibilities of producers and network operators	Intermediate
Extensive liberalization of the electricity market	Enhancing competition, price efficiency, and attracting investment	Limited market maturity, weak measurement, collection, and regulatory oversight	Linked to advanced requirements and not a direct priority	Gradual institutional and regulatory reform, clear separation of roles, and development of an effective regulatory body	Advanced
Flexible markets and advanced electricity trading	Dynamically manage supply, demand, and prices	Incomplete digital, regulatory, and market infrastructure	Specific to an advanced energy system and not a current priority for Iraq	Developing the network, measurement, market, and institutions before implementation	Advanced
Accelerated decarbonization with high quantitative targets	Rapidly reducing emissions and replacing fossil fuels	The need to address supply security, network reliability, and increasing demand continues	Not a direct priority in its German form	Adopting incremental goals linked to readiness, energy security, and investment capacity	Intermediate then advanced

Accelerated phase-out of traditional power generation sources	Restructuring the energy mix and reducing emissions	High reliance on fossil fuels and insufficient current alternatives to meet demand	It is linked to an advanced German context and cannot be directly translated	The reduction was linked to the availability of reliable alternatives, network modernization, and ensuring security of supply	Advanced
Digital participation in energy management	Empowering consumers, improving transparency, and responding to demand	Unequal access to energy-related digital technologies and services	It requires a fundamental change	Addressing the digital divide, protecting data, and ensuring equitable access	Intermediate
Advanced integration between energy and smart urban planning	Linking energy to land use, buildings, transportation, and digital infrastructure	There are opportunities for integration, but institutional and data application remains limited	It requires gradual adjustment	Incorporating energy requirements into urban plans, developing databases, and coordinating between institutions	Intermediate then advanced

The different components of the Energiewende are sorted by their flexibility towards implementation in the Iraqi energy system and mapped onto a step-by-step pathway of implementation shown in Figure 3. The phases are not intended to be set to a time frame, but they are intended to show increasing levels of readiness. To achieve progression, it is necessary to have strides in the following areas: technical and network readiness; financial and investment capacity; institutional and regulatory quality; community acceptance and participation; equity in energy access; renewable-energy potential and energy efficiency.

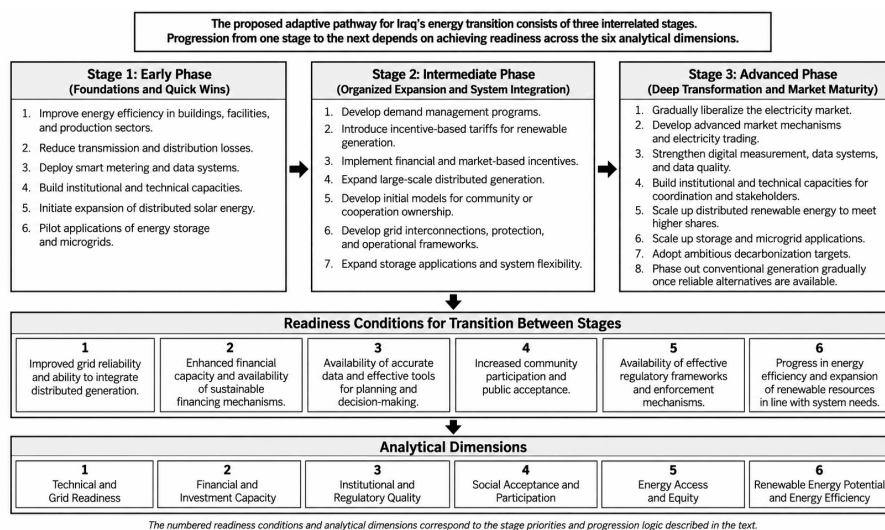


Figure 3. Adaptive multi-stage framework for Iraq's energy transition

4.3 Assessment Based on the Six Analytical Dimensions

4.3.1 Technical and network readiness

The analysis showed that technical and network readiness is one of the most influential dimensions in determining the adaptability of Energiewende elements to the Iraqi context. Indeed, grid reliability, losses in both transmission and distribution, the ageing of some of the key components of the electricity grid, and poorly developed metering and data systems constrain any potential smooth transition to higher levels of renewable generation or to more challenging electricity markets. The demand for electricity in Iraq grows at about 5%–7% per year, which exacerbates the already strained relationship between supply and demand due to shortages in electricity supply, the loss of transmission lines, and the old generation units (Altai et al., 2022).

Accordingly, loss reduction, energy-efficiency improvement, smart metering, and technical capacity building are prioritized in the early stage. While these elements would not fundamentally change the structure of the market, they are necessary ingredients to enhance the performance of the market system and prepare it for higher levels of distributed-generation integration.

Distributed solar power generation, storage, and microgrids are becoming more and more relevant, but there are many other challenges on the way to widespread application, such as the development of interconnection standards, protection systems, bidirectional flow management, etc.; these concepts will eventually be a reality, which will give the network more resilience. While these elements can be flexibly adapted, their large-scale deployment needs to be supported by phased enhancements to both interconnection standards and protection equipment, bidirectional-flow management, and grid resilience.

4.3.2 Financial and investment capacity

The analysis of the types of instruments that can be adopted and the sizes of projects that can be implemented indicated that financial and investment capacity plays a part in the selection of both. Ongoing long-term investment commitment, risk-management tools, and a more certain investment climate are the three necessary measures to underpin the deployment and storage of renewables and grid modernization.

As a consequence, financial and market incentives, incentive tariffs, community ownership, and massive distributed generation were determined to be elements that should be dramatically changed. The transfer of these instruments from Germany directly might result in financial and institutional responsibilities that the existing Iraqi instruments are not able to cope with.

Based on the output, financial adjustment should be made in an exclusively sequential manner; incentives have to be prioritized projects, financial mechanisms should be developed for the household and small business sector, the public-private partnerships should be developed, and the margins for support instruments should be tested and then scaled up.

4.3.3 Institutional and regulatory quality

The analysis identified institutional and regulatory quality as a cross-cutting precondition for the effective implementation of market, regulatory, and technical instruments. Even if renewable resources and technology are available, ensuring the successful implementation of these resources can be a challenge if institutional roles are not fully coordinated, regulations are not fully developed, and enforcement is weak. Clear responsibilities, effective oversight, and institutional coordination directly influence the capacity to implement, monitor, and scale up energy-transition projects. Fragmented responsibilities, lengthy procedures, and weak coordination among ministries, electricity-sector institutions, regulatory bodies, and local authorities can increase administrative requirements, delay approvals, and weaken project monitoring (Al-Rikabi et al., 2026).

That is why institutional capacity building is considered an early stage and an ongoing process, and market liberalization, advanced electricity trading, and large-scale community ownership are considered intermediate and advanced stages. Early-stage institutional capacity building should involve clarifying the mandate, enhancing inter-agency coordination, building staff capacity in renewable-energy projects, creating monitoring and evaluation processes, and developing technical and regulatory norms and parameters to be adhered to by renewable energy projects.

However, in addition to the lack of some laws, A further challenge is that regulatory frameworks must be enforceable in practice; there needs to be an effective regulatory body; clear mechanisms for interconnection, tariffs, and monitoring need to be in place; and there needs to be the institutional capacity for the decisions to be implemented consistently. Regulations that are not backed up by efficient monitoring/sanctioning/institutional accountability, or easy-to-follow procedures and pathways for implementation, are unlikely to have consistent impacts on investment decisions or project performance. Lack of clear and complete rules and regulations for grid connection, the purchase of electricity, tariff rates, and small-producer participation further hinders the development of rooftop solar PV or other distributed power generation. Thus, Institutional and regulatory development should proceed in parallel with technological deployment rather than follow it.

4.3.4 Community acceptance and participation

Results of the analysis revealed that community acceptance and participation remain less developed than the technical dimension; however, this is highly relevant for the success of distributed generation, community ownership, demand management and digital participation in energy management. While many widespread decentralized electricity solutions can be found, the existence of decentralized electricity does not imply that any form of organized participation, collective ownership, or mechanism exists for the communities to influence electricity project implementation or planning.

Although decentralized electricity solutions are already widespread, their presence does not necessarily imply organized community participation, collective ownership, or formal mechanisms for public influence over energy planning and project implementation. Meanwhile, the embeddedness of local generation implies that households

and businesses are already comfortable with a certain level of decentralization of the provision of their energy supply, and provides a basis for the gradual acceptance of local solar generation, storage, and microgrids. Thus, there needs to be a level of trust-building, an increase in awareness, an understanding of the costs and benefits, and equitable ways to share risks and rewards when moving to distributed solar systems or cooperative ownership arrangements. Confidence in the quality of equipment, services and maintenance, billing systems, the promises of financial gains, and the transparency of the institutions overseeing projects are likely to be crucial when bringing about public acceptance (Al-Ghabera et al., 2024). Initial investment costs and financing options are other potential barriers to acceptance, as well as the lack of evident or guaranteed savings, especially for those with lower incomes or for small businesses (Harsha, 2025). The results indicate that rather than large-scale collective ownership models from the start, it should start with local and pilot projects as well as awareness-raising activities and put in place effective consultation mechanisms. Pilot projects can offer opportunities to evaluate technical performance and mechanisms for finance, maintenance, benefit sharing, and degree of acceptance by the public before consideration of regional or nationwide replication. Local authorities, universities, civil-society organizations, and community representatives could support site selection, awareness raising, consultation, and communication of local benefits once their roles are clearly defined and coordinated with national energy institutions (Neama & Abbood, 2026).

Hence, community acceptance and participation should be considered as readiness conditions that develop gradually. The initial implementation should focus on transparency of information; financing options, as they will be cheaper; participation, whereby it is as accessible as possible; pilot projects, as these will be implemented at the local level; and short-term benefits, whereby they will be visible.

4.3.5 Equity in energy access

The analysis showed that the transition away from oil and gas should not be evaluated solely by the share of renewable energy. It should also consider affordability, equitable geographic distribution of services, protection of vulnerable groups, and access to energy-related digital technologies and services.

Geographical equity means that not only should the focus be on funding renewable energy projects with better infrastructure or higher returns on investment, but areas with weak electrical infrastructure, power outages, remote settlements, and important public services must also be involved in the initial renewable energy projects. A transition may increase renewable capacity while remaining socially unequal if it does not improve the reliability, affordability, and accessibility of energy services (Tahir, 2026).

Certain market mechanisms or incentives could be effective to benefit those who can afford to invest in solar technologies, but those who are most vulnerable are still reliant on sporadic supplies or higher-cost solutions. Households that depend on both public electricity and supplementary private generation may face overlapping energy costs, increasing the risk of energy poverty among lower-income groups (Albdiery & Al-Mosawy, 2024b).

It is therefore important that the design of tariffs and incentives does not increase inequalities in society or impose the burden of transition onto those who are less able to afford the choices arising from transition. Less digital literacy, poor internet connections, or limited access to internet-enabled devices may result in lower uptake of smart tariffs, digital billing, demand-response systems, and online devices and programs. The collection and use of consumption data also require clear rules governing ownership, access, privacy, storage, and authorized use. If implementing rooftop solar without complementary measures such as pro-poor financing and access to solar provision to tenants and low-income consumers, there is a risk that the potential benefits of rooftop solar will be unequally distributed, benefiting higher-income households and the property owners (Hassan et al., 2024). Also, factors related to the digital divide, data protection, and differential access to technologies need to be taken into account for smart metering and digital services to be expanded. This means that, to assess the quality of the transformation process, equity should therefore be treated as a design criterion rather than as a secondary outcome. To address such equity-oriented aspects, targeted subsidies are needed (Ersoy & Terrapon-Pfaff, 2021). For instance, minimum service guarantees, consumer-protection mechanisms, geographically balanced selection of projects, and participatory mechanisms can be considered.

Therefore, equity in energy access needs to be addressed as a readiness condition that cuts across from the initial phase to the end. Firstly, early interventions should focus on thinking about affordability, the reliability of basic services, and unserved areas; secondly, later, a digital/market-based approach should be considered only if measures are put in place to ensure that vulnerable consumers are not deprived of access and that costs to consumers do not unfairly increase.

4.3.6 Renewable energy potential and energy efficiency

The analysis identified this dimension as a major strength in the Iraqi context because of the country's high solar potential and substantial opportunities for energy-efficiency improvement and loss reduction. In general, the solar irradiation in Iraq ranges from about 5.5–6.5 kWh/m²/day, whereas the annual hours of sunshine duration are about 2,800–3,300 hours over that area, which represent favorable conditions for solar energy applications (both distributed and utility solar systems). Western and southern parts of Iraq have a large open space and high

irradiation values, which are the best areas for the development of solar energy. Nevertheless, renewable electricity generation—other than hydropower—remained very limited, and renewables as a share of total electricity generated are about 0.3% (recent energy-sector assessment by Aenert (2025)).

However, natural-resource potential alone is insufficient to ensure a successful transition. Thus, distributed solar energy and energy efficiency were identified as high-potential elements and connected to the phased implementation and explicit technical and financial conditions. Distributed Solar systems can aid in providing early benefits through the supply of local loads and reducing pressure on the grid load; however, the large-scale deployment of distributed solar production demands standards for installations, accurate metering, financing mechanisms, installation maintenance capacity, and clear rules for interconnection. The small proportion of renewable energy in Iraq's electricity supply, even though Iraq has vast solar potential, means that solar resources have not yet resulted in a similar proportion of installed capacity or electricity generation (Al-Kayiem & Mohammad, 2019).

An important finding of the results is that improving energy efficiency and loss reduction should go hand in hand with or precede the addition of renewable generation, as the investment needed is reduced, the pressure on the existing system is alleviated, and the amount of benefit the renewable generation is able to deliver is enhanced. Energy efficiency investments can also be considered an immediate system-benefit measure since avoiding electricity consumption and technical losses frees up electricity supply without having to invest in an equivalent one. Another piece of evidence of the spatial distribution of solar irradiation indicates that significant parts of Iraq, mainly the western and southern ones, are suitable for solar energy development, as shown in Figure 4.

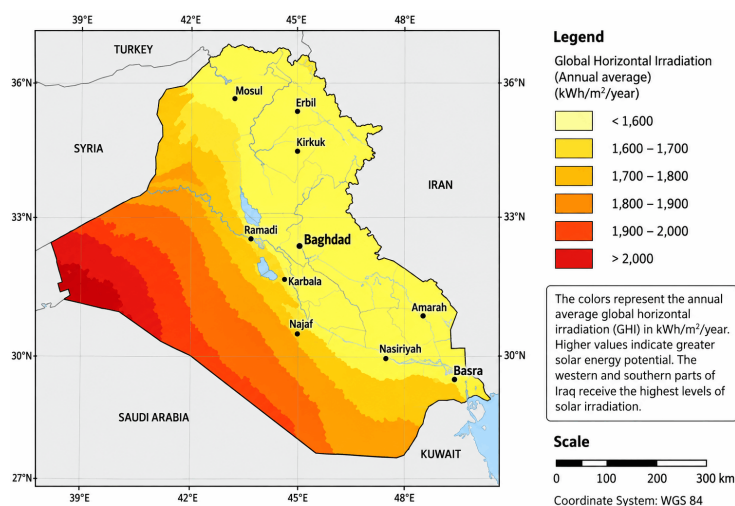


Figure 4. Spatial distribution of global horizontal irradiation in Iraq (long-term average for the period 1999–2015)

Source: World Bank and Solargis.

In this regard, renewable-energy potential and energy efficiency have the best technical argument to start Iraq's transition, while, to be really effective, there should be phased implementation. In the first stages, bigger renewable generation and deeper system integration should be phased in in parallel with the increasing readiness of the grid, finance, regulation and institutions in terms of efficiency and loss reduction, and distributed solar applications and pilot storage systems should be prioritized.

4.3.7 The integrated outcome of the six dimensions

The six-dimensional assessment shows that the adaptability of Energiewende elements depends on the interaction among technical, financial, institutional, social, and equity-related conditions rather than on any single dimension.

Based on the above, adaptability should therefore not be treated as an inherent or fixed property of an energy-transition element. Many items of the same element may be appropriate for early (e.g., limited) adoption given the current situation but may need significant institutional, financial, or technical development for any further rollout.

Thus, the dimensions should be interpreted as enabling or constraining conditions that influence when, how, and to what extent each transition element can be implemented. Renewable-energy potential and energy efficiency are the main opportunity base, and the technical and network readiness, institutional and regulatory conditions, and financial capacity are the major obstacles for implementation. Technically viable interventions can only become socially legitimate and inclusive if they have the backing, acceptance, and equity that are derived from adequate energy access among communities (Deed et al., 2025). The integrated assessment also indicates that a severe constraint in one dimension can become a critical bottleneck, even when other dimensions are supportive.

High solar potential cannot compensate for the absence of interconnection standards, metering, financing, and institutional enforcement. Similarly, incentives that are offered to someone are difficult to achieve strong results when they are not able to be added to and monitored by the grid and regulatory system. Based on this, the following priorities are mapped out in the integrated priority sequence: optimization of system efficiency, reduction of losses, development of metering and data systems, and institutional and technical capacity building. These measures provide the foundation for a progressive rollout of distributed solar energy, storage, microgrids, demand management, and targeted financing options. More in-depth market reforms, significant community ownership, and higher quality electricity trading should only be implemented once the respective readiness conditions are significantly enhanced.

This is because some elements move from initial non-scalable “pilot” implementations to more comprehensive and sophisticated implementations. The results therefore do not recommend starting with instruments most technologically advanced, or those with the highest political end. Rather, implementation should start with elements which address some of the gaps in the system and which pave the way for its future transformation. There are many cross-dimensional enabling early-stage interventions. Their benefits are step-by-step reduction in network losses leading to better technical performance and financial efficiency, smart metering for better monitoring and billing, demand management and tariff implementation, institutional capacity building for regulation, investment and project oversight, and local pilot projects to create evidence and build public trust and learning. The combined result of the six analytical dimensions presented in Table 5 shows that the adaptability of energy transition elements in Iraq not only relies on resource endowment but also on the interaction between stage of technological readiness, level of financial capability, institutions’ qualities, and degree of social acceptance and equality of access to energy. Also indicated in the table are some of the dimensions that lie more in the back end of opportunity available (e.g., renewable energy potential and energy efficiency), and those that are more prerequisites for implementation and expansion (e.g., agricultural land).

The results show that the six dimensions of IGR are not mutually exclusive, but are opportunities, constraints, and readiness conditions that are interrelated. A very good solar potential combined with a very poor electricity grid, for example, will not lead to sustainable development if not supported with regulations, really good financing schemes, a positive attitude by society, and promises of equality of benefits and costs. Accordingly, it needs to start with actions that tackle fundamentals as well as set the stage for moving on to higher stages of transformation.

Table 5. Integrated assessment of the six analytical dimensions and their implications for the energy transition in Iraq

Analytical Dimension	The Current Situation in Iraq	Key Opportunity or Constraint	Enabling Requirements	Priority Phase
Technical readiness and network readiness	The electricity system suffers from poor supply reliability, high transmission and distribution losses, outdated parts of the infrastructure, and limited advanced measurement, monitoring, storage, and interconnection systems.	Weak grids represent one of the most significant constraints to the organized expansion of renewable energy and distributed generation, but at the same time they provide an opportunity to link the energy transition to a comprehensive system modernization program.	Reducing losses, rehabilitating transmission and distribution networks, expanding smart meters, improving data collection, setting clear standards for network connectivity, and implementing pilot projects for storage and microgrids.	Early and continuous
Financial and investment capacity	The sector faces limitations in public funding, weak investment guarantees, high risks, and unstable pricing and return mechanisms, along with poor access to appropriate financing for households and small businesses.	Lack of funding limits the ability to expand projects, while diversifying financing tools and attracting private investment provides an opportunity to accelerate implementation and reduce the burden on the public budget.	Concessional financing, investment guarantees, public-private partnerships, risk-sharing mechanisms, and financing programs focused on distributed solar energy and energy efficiency.	Early to intermediate

Institutional and regulatory quality	The institutional environment is characterized by overlapping responsibilities, incomplete regulatory frameworks, weak coordination, follow-up, and implementation, and the absence of unified and clear procedures for licensing, connection, tariff, and electricity purchase. There is extensive community experience in relying on decentralized solutions such as private generators, but models of organized participation, community ownership, and consultation mechanisms are still limited.	The limited institutional capacity represents a comprehensive constraint affecting technical, financial, and social aspects, but at the same time it is an area where reform can have a broad enabling effect.	Clarifying jurisdictions, enhancing coordination between institutions, building technical and administrative capacities, establishing enforceable rules for linking, tariffs, and monitoring, and promoting accountability and transparency.	Early and continuous
Community acceptance and participation	Service reliability and costs vary between social classes and regions, and some families bear multiple burdens as a result of relying on the public network, generators, and home solutions, with a digital divide affecting the utilization of smart services.	Weak trust, awareness, and affordability may limit the acceptance of new technologies, while familiarity with local generation can be used to facilitate the acceptance of distributed solar power and microgrids.	Awareness campaigns, transparent cost-benefit information, local pilot projects, clear consultation mechanisms, accessible participation options, and assurance of equipment quality and maintenance services.	Early to middle
Equity in access to energy	Iraq has high solar energy potential and great opportunities to improve consumption efficiency and reduce losses, but the contribution of renewable energy to the system is still limited compared to the size of the available resources.	Untargeted incentives and programs may widen inequalities, whereas transformation can improve equity if it prioritizes the least served groups and regions.	Targeted support for low-income groups, affordable financing, guaranteed minimum service, balanced geographical distribution of projects, consumer and data protection, and provision of alternatives for those unable to access digital technology. Prioritize energy efficiency and loss reduction, expand distributed solar applications, establish clear technical and financial standards, and link large-scale expansion to improved grid capacity, storage, and financing.	From the early stage and through all stages
Renewable energy potential and energy efficiency		Solar energy and energy efficiency represent the strongest technical basis for initiating the transformation, but their actual exploitation depends on grid readiness, financing, regulation, and institutional capacity.		From early to advanced gradually

4.4 Multi-Level Perspective Interpretation of Iraq's Energy Transition

The results of the contextual assessment and adaptability classification are interpreted using an interpretive framework—the Multi-Level Perspective. In Iraq's context, the energy transition is conditioned by a complex set of landscape dynamics as well as the stable character of the key socio-technical regime, and the emergence of niche innovations in and around the electricity system (Ersoy & Terrapon-Pfaff, 2021).

4.4.1 Socio-technical landscape pressures

At the socio-technical landscape level, external pressures such as those faced by Iraq's energy system are present and cannot be manipulated by single players in the electricity sector. Pressures involve population and urban growth, rise in electricity demand, cooling demand, environmental pressures due to climate change, energy-

security concerns, international decarbonization commitments, and also reduction in renewable-energy technologies cost and increasing adoption of renewable energy (Hassan et al., 2024). However, all these pressures lead to a heightened sense of the need for change, while the rate and direction in which change occurs also depends on the alternatives that are available and that are themselves able to transition.

4.4.2 The existing socio-technical regime

The socio-technical regime is the current dominant electric power generation mode relying on fossil fuels, and the institutional setup is rather unchanged. Analysis of indicators shows that this is a highly rigid system to change, and it is linked to the economic/political system. The regime has an impact on how things are produced, what technology is used, how it is organized in terms of its policies, what the market structure is, and how it is being used. It is a multi-part system, which consists of several components: Industry, Policy, Technology, Culture, Science & Scientific Research, and Markets & User Preferences.

The electricity system in Iraq is traditionally characterized by centralized generation driven by fossil fuels, clearly demarcated institutional powers and duties for the different roles, a non-market price regulated by the State, low performance of the electricity system, weak legal attitude toward DG, and high reliance on electricity from the State and private generation. These technical, institutional, financial, and behavioral setups extend and strengthen each other, and help to consolidate the current system. Moreover, the system of electricity generation, its cost, institutions, investments, and consumer habits are all geared towards the current regime and make a rapid transition impossible.

4.4.3 Niche innovations and protected experiments

Niche innovations represent spaces in which alternative technologies and practices can emerge, be tested, and gradually develop. Their characteristics include limited research networks, small-scale experiments, innovative technologies, and experimental investments. Relevant indicators may include the number of start-ups, pilot projects, experimental technology prototypes, and the scale of innovation funding. All of these characteristics have quantitative measurement indicators (number of startups, number of pilot projects, number of experimental technology prototypes, size of innovation funding). New choices at the niche level involve rooftop solar power, distributed generation, energy storage, microgrids, smart metering, demand-management applications, energy-efficiency projects/strategies, and local demonstration projects. Most of these options remain limited in scale and lack sufficient regulatory, financial, institutional, and technical support to challenge the dominant regime or provide widely accessible alternatives (Hamid, 2025; Yahia et al., 2026).

Their immediate significance is not the potential to replace the national electricity system but more the opportunities to provide technical evidence for alternatives, generate implementation experience, establish networks of actors, finance and maintenance of alternatives, and raise awareness of institutions and the population for the use of alternatives.

4.4.4 Interaction between the three levels and the phased transition pathway

Transition potential emerges from the interaction among the three levels rather than from developments at any single level. This need for change increases due to landscape pressures like supply insecurity and a widening supply-demand gap, or climate stress and international decarbonization expectations. But niche innovations spread and are limited by regime restrictions such as weak grids, institutional weaknesses, financing, and incomplete regulations. Niche innovations can more easily grow under two conditions: when they are directly addressing the limitations of regimes and when conditions resemble ‘supportive institutional reforms’ which allow them to be tested and learnt, and then gradually scaled. Staged interventions of distributed solar energy, storage, microgrids and smart metering are therefore more likely as interventions than as immediate replacements for the whole centralized energy system. This interaction is summarized in Figure 5 by presenting three different elements affecting the most probable pathway of transition in Iraq.

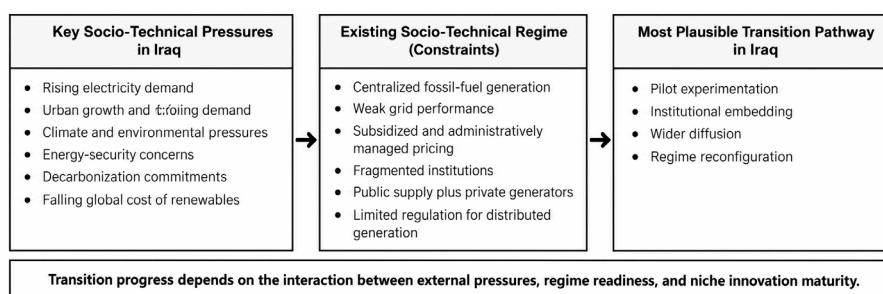


Figure 5. Multi-level interpretation of Iraq's phased energy transition

Source: Developed by the authors based on the Multi-Level Perspective and the findings of the contextual and adaptability assessments.

When the niche innovations are in the early phase, they are expected to coexist with the established regime and have complementary roles. Some of these innovations might grow and be incorporated into the regime's mainstream practices—along with the growth in grid reliability, regulation, institutional capacity, financing, and social acceptance, they may eventually bring about a transformation of the regime itself. As illustrated in Figure 5, the results indicate that there is no full regime replacement in Iraq today. Rather, it seems that the much more likely pathway is a gradual reconfiguration process, in which incremental additions of niche innovations to the existing system are used to improve system performance, address supply deficiencies, and set up conditions for more substantial structural change.

Consequently, the proposed phases should not be interpreted as fixed time periods, according to the Multi-Level Perspective. Changes in the relationship among landscape pressures, regime readiness, and niche maturity are required for progress. This early phase aims for protection and testing of appropriate innovations with a focus on filling gaps in the existing regime; the intermediate phase aims for scaling up successful innovations alongside institutional, financial, and grid reforms; and the advanced phase becomes possible when innovations, so far considered marginal, have reached a sufficient level of maturity and are introduced on a broad scale.

Thus, Multi-Level Perspective is not a distinct theory level to be added to the analysis. It facilitates linking the contextual assessment, Energiewende elements classification, the six readiness dimensions, and the meshing of the proposed adaptive pathway.

4.5 Phased Adaptive Framework for Iraq's Energy Transition

The study has presented a phasing (adaptive) framework for Iraq's energy transition based on the above-mentioned contextual assessment, the Energiewende elements classification, the six analytical dimensions and the Multi-Level Perspective interpretation. The aim of the framework is to be more modular, flexible, and adaptive, not to emulate the German transition pathway. It arranges flexible and customized components in view of the state of readiness in Iraq at the time and restrictions on implementation.

Stages 1, 2, and 3 refer to early, intermediate, and advanced, and do not necessarily require spending time that equals that stage, or arbitrary numbers. They represent respective increases in the level of technical, institutional, financial, regulatory, and social readiness. The transitions between the stages will be based on observable improvements in the system performance and capacity to implement, and not on the timing of the transition.

4.5.1 Early stage: System Stabilization and Enabling Foundations

The early stage is designed to build the minimum enabling conditions for transition, and to start to tackle gaps in the existing electricity system. These six priority areas can be considered as measures of intervention and include loss reduction in transmission and distribution networks, rehabilitation of critical infrastructure in the grid, expansion of metering and data systems, improvement of billing and monitoring, strengthening of institutional coordination, and establishment of clear procedures for the licensing and grid connection of renewables.

The primary purpose of this first phase, therefore, is not focused on 'decarbonization', but on making the system more stable and increasing institutional learning, and creating credible implementation mechanisms. The key actions, with the parties involved, readiness criteria and expected outcomes for this step are outlined in Table 6.

4.5.2 Intermediate stage: Expansion and Institutional Embedding

At the intermediate stage, the critical regulatory process, technical standards, funding systems, institutions and other monitoring mechanisms to support expansion have been established. Successful pilot projects in distributed solar, storage and microgrids, smart metering and demand management can be scaled up and more systematically embedded in the electricity system at this point. This expansion should be accompanied by focus on financing, risk-sharing facilities, public-private partnerships, enforceable institutional responsibilities, which are enforceable, and monitoring of technical, financial and social outcomes.

4.5.3 Advanced stage: System Integration and Structural Transformation

A transition to the advanced phase is possible only if there are long-term improvements within the realms of grid reliability, regulatory effectiveness, institutional accountability, financial viability and public acceptance within the electricity system. It is at this juncture that large-scale integration of renewables, extensive demand-response systems, new participation from communities, broader community ownership, extensive digitization and more sophisticated electricity market instruments are a real possibility. The introduction of these instruments should continue to be subject to the technical and institutional ability of the Iraqi system to effectively operate, monitor, and regulate them. Table 6 provides an overview of priority actions, actors responsible, readiness indicators, and expected outputs of the three stages in the proposed adaptive framework.

4.5.4 Progression conditions and the adaptive character of the framework

One's readiness for each change of stage must be discernible. These include: improvements that can be

measured in the reliability of the grid and the reduction of losses; functioning interconnection and licensing procedures; effective metering and billing systems; sustainable financing is available; pilot projects have been successfully executed; coordination of the institutions is addressed; and safeguards for participation and equitable access and for affordability are demonstrated. When these conditions are not met, this does not automatically stop transition activity; transition activity may only be expanded on certain instruments in an appropriate manner as part of transition activities to a limited scale, in limited complexity and speed.

Table 6. Phased implementation framework for Iraq's adaptive energy transition

Priority Action		Main Entity	Supporting Entities	Readiness or Achievement Indicator	Expected Output
Early stage	Ministry of Electricity	Transmission and distribution companies and regulatory bodies	A documented decrease in losses and improved supply reliability	Increase available electricity and improve network efficiency	Implementing a program to reduce transmission and distribution losses
	Ministry of Electricity	Ministry of Planning and Funding Agencies	Completion of rehabilitation projects in priority locations	A network more capable of accommodating distributed applications	Upgrading critical network components
	Ministry of Electricity	Distribution companies, universities, and technology providers	Operating systems for measuring and collecting verifiable data	More accurate data for billing, losses, and demand management	Implementation of smart metering in pilot areas
	Ministry of Electricity and the competent regulatory body	Legal authorities and governorates	Issuance of written, standardized, and applicable procedures	Reducing ambiguity and project delays	Standardizing licensing and grid-connection procedures
	Ministry of Electricity and Training Institutions	Universities and specialized organizations	Implementing training programs and clearly defining responsibilities	Enhancing the capacity for implementation, follow-up, and evaluation	Building institutional and technical capacities
	Ministry of Electricity and Governorates	Universities, the private sector, and local authorities	Operating projects with a documented performance evaluation system	Providing practical experience before expanding	Launching selected pilot projects
Intermediate stage	Ministry of Electricity	Governorates, the private sector, and universities	Proven success of projects technically, financially, and operationally	Transition from experimentation to organized expansion	Expanding successful pilot projects
	Ministry of Electricity and the regulatory body	Distribution companies, investors, and local authorities	Applying standards for connection, protection, and measurement	Increased reliable local generation	Expanding distributed solar power and microgrids
	Ministry of Finance and Financial Institutions	Ministry of Planning, Banks and the Private Sector	Availability of loans, guarantees, and sustainable financing mechanisms	Reducing investment risks and expanding participation	Developing financing tools and sharing risks
	Ministry of Electricity and the regulatory body	Ministries, governorates, and local institutions	Clarity of powers, follow-up procedures, and enforcement	Reducing institutional conflict and delays	Strengthening institutional responsibilities

	Distribution companies	Technology providers and large enterprises	Reliable operational data and verifiable demand-response programs	Improved billing, planning, and load reduction	Expanding smart metering and demand management
	Regulatory and local authorities	Community organizations and universities	Indicators of ability to pay, access, and participate	Preventing unfair expansion of benefits and costs	Monitoring social outcomes and equity outcomes
Advanced stage	Ministry of Electricity	Ministry of Planning, investors, and network operators	Proven network capacity for absorption and balance	Increasing the share of renewable energy without harming stability	Large-scale expansion of renewable energy
	Distribution companies and the regulatory body	Major consumers and technology providers	Reliable measurement, applicable tariff, and verifiable response	Reducing peak loads and improving the balance of supply and demand	Implementing advanced demand response systems
	Regulatory body and local authorities	Banks, cooperatives, and local communities	Clear frameworks for ownership and the sharing of benefits and risks	Expanding local participation and equitable distribution of benefits	Expanding community ownership of energy
	Ministry of Electricity and Digital Entities	Municipalities, distribution companies, and technology providers	Interconnected systems and standards for data protection and cybersecurity	More integrated management of distributed energy and resources	Deepening digitalization and data integration
	Independent regulatory body	Ministry of Electricity, network operators, and investors	A mature market, a clear separation of roles, and effective oversight	Improving flexibility, efficiency, and investment	Developing advanced electricity market tools
	Ministry of Planning and Local Authorities	Ministry of Electricity, Transport and Communications	Shared databases and stable institutional arrangements	Integrating energy into urban plans and infrastructure	Enhancing the integration of energy and urban planning

The framework has an adaptive instead of a strictly linear approach. Varying geographic or institutional contexts or technologies may come to fruition in different ways and at varying speeds; some interventions may have a multi-stage effect. These factors—including energy efficiency, institutional capacity building, equity safeguards, and upgrades to data systems—have relevance across the entire transition process, and can manifest in different ways and at different levels of sophistication as the transition process unfolds. The framework can thus be revised, appreciated, and replicated, with feedback and overlap, as a result of the experience gained from its implementation and as the local context changes.

4.6 Smart Urban Planning Implications of the Phased Energy-Transition Framework

The proposed energy-transition framework is operationalized across multiple spatial scales, from the city to the building level, within the context of smart urban planning, which supplies the spatial, infrastructural, and governance aspects under which it is supposed to be implemented (Esfandi et al., 2024). In this work, smart-city technologies are not considered as a separate development goal nor substitutes for institutional reform. Rather, they are treated as enabling tools, which will facilitate improvements to energy-system visibility, coordination of distributed resources, demand management and the connection of energy decisions with broader urban-development priorities. The phased framework sets out when specific energy-transition instruments can become feasible, and smart urban planning determines where and how they can be translated into spatially differentiated and operational interventions.

4.6.1 Integrating energy transition into urban, neighborhood, and building planning

Energy-transition objectives should be integrated into urban-development planning and not just in the electricity sector through national policies. Land-use plans, neighborhood-development strategies, building regulations, infrastructure-investment programs and urban-redevelopment projects can impact electricity demand, renewable-energy deployment, network requirements and access to low-carbon energy services. Designing such an

integration involves planning for the right locations for distributed generation, energy storage, microgrids and energy-efficient redevelopment, taking into account current land uses and infrastructure capacity as well as environmental conditions, public-service needs and social vulnerability (Hsu et al., 2023; Schönberger & Reiche, 2016). Thus, urban plans should also include energy-related spatial indicators, such as building density, roof availability, land-use intensity, electricity-demand concentration, infrastructure condition, access to public facilities, and the location of underserved communities. These indicators can be used to prioritize interventions and should prevent energy projects from being selected only due to technical potential or investment returns.

Rooftop solar systems are one of the most readily adaptable applications at the building and neighborhood levels, as rooftops are available in urban areas and can meet local demand, and potentially help in reducing reliance on centralized generation systems. Still, they can only be used in certain building types, depending on a variety of factors, such as building form, roof area, structural condition, ownership arrangements, shading, access to financing, and electricity-demand profiles. Therefore, rooftop-solar programs must be linked to the building inventory and solar-potential assessment, strict building structural and safety demands, transparent rooftop-solar installation requirements, and differentiated financing of rooftop-solar installations for public institutions, enterprises, and households (Shi et al., 2026). The location and choice of the early stage should depend on several factors, such as schools, hospitals, universities, municipal buildings, and other public facilities. It must have a visible public benefit and be a location at which monitoring, institutional learning, and replication can take place.

Urban energy efficiency should not only be focused on equipment upgrades, but also on the energy efficiency of urban form, of building design, of land-use distribution and of urban services organization. In hot climate cities, factors like building orientation, shading, insulation, and ventilation requirements, development density, street designs, vegetation and the placement of urban activities are significant drivers of cooling demand and energy consumption in cities (Amado et al., 2016; Umoh et al., 2024). Lightly developed planning interventions—such as improved building-energy standards, new public-space buildings with enhanced thermal performance, passive cooling in new development, retrofitting of existing inefficient building stocks, and aligning energy-efficiency requirements with urban-redevelopment programs – can be realized in an early stage in Iraq. These are in line with the priority measures identified in the early stage of the framework, as they can lower the demand and enhance the performance of the system when future generation is still being added or before new additions in the generation system.

4.6.2 Smart infrastructure, microgrids, and urban energy management

Microgrids can ensure the reliability of critical urban services, particularly if the reliability of central-grid service is called into question. It is best to focus their application in the early stages on obvious urban areas or areas with important facilities where the generation, storage, demand, maintenance, and operating boundaries are clearly defined (Lin et al., 2026). These may be applied to hospitals, emergency, water-pumping and treatment facilities, university campuses, complex buildings for public services, industrial clusters, remote and less-served settlements (Ottenburger et al., 2024). These should not be thought of as one-off bits of technology. Their effective implementation will include land use planning, infrastructure management, emergency planning, financing, maintenance plans, and future needs for interconnections on the grid. In the early stage of the development of microgrids, these microgrids can be protected pilot experiments to test the technical performance, the institutional coordination, and the operational responsibilities. The successful model could be scaled up and increased to connect more systematically to larger distribution networks in the intermediate phase.

More information can be gathered, such as consumption patterns, contribution of distributed resources, peak demand, losses, and network performance, from smart metering and urban energy-data systems. This data is useful at the urban planning level in prioritizing urban infrastructure, forecasting energy demand, energy efficiency initiatives, and identifying areas with poor or imbalanced energy services (Ponnusamy et al., 2021). Demand-management applications for the future might start with large public-construction facilities, commercial buildings, cooling loads, street lights, and other electricity-consuming urban services, in which electricity usage can be measured and managed to a greater extent.

However, to fully attain the potential gains of data-driven energy management, interoperable technical systems and well-defined institutional responsibilities, data-quality standards, cybersecurity protection measures, and energy-consumption data ownership, access, data storage, data privacy, and authorized use rules are required (Pourmirza et al., 2026). Without these put in place, smart technologies could lead to a lack of integration in information systems, incomplete institutional learning, and loss of public trust. In conclusion, it is necessary to go beyond the mere technological aspects of smart metering and digital platforms and take them as one of the many components of a package of regulatory and institutional changes.

4.6.3 Local governance and institutional coordination

An urban energy transition needs to be coordinated by national electricity institutions, planning authorities and governorates, the municipality, the electricity provider, the University, private investors, civil-society groups and community representatives (Melica et al., 2018). Strategic energy policy, big infrastructure projects and system-

wide regulation are primarily the responsibility of national institutions.

There are other touchpoints for local governments to integrate energy into plans such as building-control procedures, public-service investments, redevelopment programs and land-use plans. Universities can support in terms of resource assessment, technical evaluation and training as well as monitoring and independent evaluation of pilot projects. The private sector can offer finance, installation, maintenance and technological expertise, and community groups can offer support through awareness and consultation and identification of local need.

The key to effective implementation is that these responsibilities should be clearly defined, decision-making processes are transparent, and good coordination exists. It is therefore important that there should be some local experimentation with a coordinated national framework that will allow cities and institutions to experiment with solutions better suited to their local needs without developing incompatible technical standards or uncoordinated electricity schemes.

In particular, there is a big need for institutional coordination when working the way from pilot implementation to wider diffusion. Documenting technical and social achievements, standardizing procedures, transferring knowledge, adapting successful models to other locations, and setting up a mechanism to support the various projects. Can be used to support institutional learning to prevent pilot projects from being isolated demonstrations by offering a minimum impact in the longer term (Xie et al., 2026).

4.6.4 Spatial equity and inclusive smart-energy planning

Investment in smart urban-energy planning should not be restricted to high-income neighborhoods or megaprojects or neighborhoods already characterized by a fairly advanced infrastructure. Outage frequency, weak service, and energy burden at household level, as well as accessibility of households or small businesses to participate in renewable-energy programs, digital exclusion, and access to critical facilities should also be included among the consideration factors for spatial prioritization (Nguyen & Batel, 2021; Payakkamas et al., 2023).

Without accompanying measures, programs which rely on rooftop solar power, on smart tariffs, on digital billing, on demand-response systems or similar measures, would have the potential to benefit disproportionately to the property owners, high-income families, and digitally connected persons. In addition, using targeted funding, concessional loans, shared or community-solar arrangements, investing in communities where solar facilities are lacking, developing alternative approaches to delivery of goods/services beyond digital technologies, consumer protection mechanisms, and implementation strategies to prevent disproportionate financial burden on vulnerable consumers are all examples of inclusive implementation options (Phelps & Lanza, 2025).

But early projects have to be targeted, too. In early projects, the site with the most significant social improvements that can be achieved through energy interventions and where there is spatial equality should also be a consideration for site selection. This could include areas having an underdeveloped grid, far-flung communities, low-income neighborhoods, hospitals, schools, water supply facilities, and other critical community facilities. In addition to technical viability and financial return, therefore, reliability issues, social vulnerability, and the distribution of the expected benefits should be included as criteria in deciding on a choice of projects.

However, it should be comprehensively addressed too: the digital aspect of smart-energy planning. Digital services (consultation, demand and customer management, smart tariffs and digital billing) may not be easily available to everyone, as the provision of these services needs digitally competent consumers, good internet access and compatible devices. Therefore, data-driven systems need to offer accessible alternatives and clear communication, as well as privacy protections and clear procedures for challenging decisions related to billing or services (Campana et al., 2025).

The siting, beneficiaries, financial, tariff, and administration of digital-service systems should all be based on equity. In this way, disparities in access to energy services due to spatial, social and digital factors can be minimized, and their reliability and availability can be enhanced.

Smart urban planning, overall, is the execution of the phased energy-transition framework into various steps that can be realized in space and implemented. Targeted distributed-solar applications, basic metering systems, critical-service microgrids, public-building projects, and energy efficiency can be the target of urban actions in the early stage. The intermediate stage can foster scaling up successful neighborhood- and district-level systems, incorporate energy data into the planning process, strengthen financing mechanisms, and expand community participation. Pathways from the existing applications to more complex ones such as smart-grid, electricity-market, and community-ownership applications should be built up using regulatory, institutional, technical, financial, and social readiness.

Smart technologies are therefore a facilitating factor for the transition, but not a substitute for grid rehabilitation, institutional reform, financial capacity, or equitable access. They rely on their success in integrating with spatial planning, infrastructure construction, and local government and social-protection goals.

4.7 Study Limitations and Future Empirical Validation

The results of this study must be seen within the context of some methodological limitations. This is a

descriptive study which is mainly based on secondary data, policy documents, institutional reports and published academic literature. While these information sources offer an adequate foundation to make a comparative and contextual evaluation, they cannot fully substitute for primary empirical evidence that can only be collected from energy-sector institutions, local authorities, investors, service providers, and electricity consumers. There were no interviews, stakeholder workshops, household surveys, or expert elicitation specifically used for the validation of the proposed classification and phased framework. In addition, it did not perform power-system simulations, techno-economic optimization, cost-benefit analysis, scenario modeling, or quantitative forecasting of renewable-energy penetration. Accordingly, the proposed stages should not be interpreted as predictions of capacity expansion, investment requirements, emission reductions, or implementation timelines. The readiness conditions were not converted into fixed numerical thresholds as the available secondary data come from varying reporting years, institutional sources, geographical coverage, and indicator definitions, and consistent evidence is not yet available across the technical, financial, institutional, social, and equity dimensions. The framework fails to reflect the differences between the performance of the electric grid in governorates, cities, neighborhoods, and rural settlements, in addition to the institutional capacities in these settings, the electric power-generating potential of renewable sources, household income, dependence on private generators, and availability of digital services, as they were all assessed at the national level. In addition, the comparison between the *Energiewende* in Germany and that of Iraq is not one of like-for-like, and the research is not interested in finding a transfer of equivalence between the *Energiewende* in Germany and in Iraq, but rather a transfer of functions and mechanisms. In a similar manner, the use of the Multi-Level Perspective is not a complete (longitudinal) reconstruction of the socio-technical transition in Iraq, but rather an interpretive make-up. This should then be complemented with additional studies to test the framework with stakeholder interviews, expert assessment, a household/business survey, as well as detailed city- or governorate-level case studies. In addition to the power-system modelling, spatial suitability analysis, techno-economic assessment and evaluation of pilot projects are also additional steps for validation. Long-term monitoring would also be required to measure the impact on and effectiveness of improvements in grid operation, coordination between institutions, investment confidence, public acceptance and equity of access, following early interventions. The framework can thus be regarded as a proposal which is based on analysis and should be further tested and refined by new data, experience on implementation and institutional changes.

5. Conclusions

In this research, comparative and context sensitivity analysis frameworks were applied to analyze the possibility of applying selected *Energiewende* principles in Iraq's energy and urban context. The German case was not seen as a model to be directly transferred but as a combination of highly adaptable and other elements which needed significant moderation, as well as elements which required more advanced institutional, technical, financial and market structures. These results show potential for Iraq in the following areas: energy efficiency, distributed solar PV, smart metering, institutional capacity development, transmission losses, distribution losses and pilot-level storage and microgrids. Other features of the market—such as more advanced market liberalization, high shares of large-scale community ownership, complex electricity trading arrangements, and aggressive decarbonization targets—require more fundamental changes and should not be considered until there have been substantial improvements in reliability of the electricity grid, the effectiveness of all the other facets of the regulations, funding capacity, institutional coordination and public acceptance.

For all six of the analytical dimensions, dealing with the transition instruments is not a stand-alone approach. The key opportunity base is the renewable-energy potential and energy efficiency, while technical and network readiness, financial and investment capacity, and institutional and regulatory quality are the main constraints on implementation. Developing technically viable interventions does not ensure socially legitimate outcomes and inclusion of energy access—community acceptance, participation, and equity of how energy is accessed are vital. These insights lead to a flexible and evolutionary approach, rather than fixed timelines or the immediate adoption of advanced policy instruments.

The Multi-Level Perspective also suggests that the situation in Iraq is not yet a complete regime replacement. The slower, more likely to happen, gradual reconfiguration is the introduction of niche innovations such as distributed solar power, storage, microgrids, smart meters, and demand management at first as complementary technologies to an existing electricity system; they create experience in the operation and implementation of those technologies; they address the urgent system needs; and over time, they become part of the electricity system. That is, the more integrated they are, and the more the existing regime gradually evolves, the more that can happen as regulatory and institutional readiness, financial, technical, and social readiness improves and innovations are found to be successful and more are adopted into the existing regime.

It therefore seeks to begin with stabilization of the system, cutting the losses and enhancing energy efficiency, enhancing metering/data systems, institutional changes, and pilot projects with care. By increasing financing, enhancing regulation, streamlining implementation, and increasing participation, the intermediate stage enhances successful applications. The advanced stage provides the means to have more renewable energy integrated, more

complex market conditions, to have communities more widely engaged in ownership, and for the most extensive digitalization, but only under readiness conditions. This is not a linear process—in some cases, technologies, institutions, and locations can overlap, and there may be feedback and varying speed of development.

This research shows that with smart urban planning, the spatial and operational environment in which the pathway is implemented is provided. By implementing spatial-equity criteria, energy-data systems, smart metering, microgrids, rooftop solar systems, public-facility programs, urban plans, building regulations, and city, district, neighborhood and building-level interventions, there are ways to connect national energy-transition goals to spatially specific measures. Smart technologies should then be seen not only as tools to help energy management, distributed generation and distributed demand reduction, but also as enabling instruments, and not as substitutes for grid rehabilitation, institutional reform, financing capacity or equitable access.

The most important contribution of the study is the adaptive, analytical framework that integrates Energiewende principles, the Multi-Level Perspective, six readiness dimensions and smart urban planning into a phased pathway specific for the context of a country with fragile grid performance, institutional and financial constraints, as well as varying degrees of energy access. The document is a framework and does not precisely predict or provide a blueprint for implementation. Instead, it provides a framework on which to work to complete a prioritization of interventions, a sequence of “reform steps”, and to explore when certain transition instruments might be achievable. This framework must be tested and enhanced in various cities and regions of Iraq with the participation of stakeholders, spatial analysis, techno-economic assessment, pilot projects, and monitoring over the long-term.

Author Contributions

Conceptualization, K.A.A.-S. and A.M.H.; methodology, K.A.A.-S. and A.A.A.-J.; software, K.A.A.-S.; validation, K.A.A.-S. and A.M.H.; formal analysis, K.A.A.-S. and A.A.A.-J.; investigation, A.M.H.; resources, A.A.A.-J.; writing—original draft preparation, K.A.A.-S.; writing—review and editing, A.A.A.-J.; visualization, A.A.A.-J.; supervision, K.A.A.-S.; project administration, K.A.A.-S.; funding acquisition, K.A.A.-S. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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Appendix

Table A1. Evaluation of energy transition elements across the six analytical dimensions

No.	Energy Transformation Element	Technical and Network Readiness	Financial and Investment Capacity	Institutional and Regulatory Quality	Community Acceptance and Participation	Equity in Access to Energy	Renewable Energy Potential and Energy Efficiency
1	Improving energy efficiency	S	CS	CS	CS	CS	S
2	Reducing losses in electricity transmission and distribution	S	CS	CS	S	S	S
3	Distributed solar energy	CS	CS	CS	CS	CS	S
4	Electricity demand management	CS	CS	CS	CS	CS	S

5	Smart measurement and data systems	CS	CS	CS	CS	CS	S
6	Building institutional and technical capacities	S	CS	S	S	S	S
7	Pilot storage projects	CS	C	CS	CS	CS	S
8	Microgrids	CS	CS	CS	CS	CS	S
9	Incentive tariff for renewable energy production	CS	C	C	CS	C	S
10	Financial and market incentives	CS	C	C	CS	C	S
11	Community or cooperative ownership of energy	CS	C	C	CS	CS	S
12	Large-scale decentralized generation	C	C	C	CS	CS	S
13	Extensive liberalization of the electricity market	C	C	C	CS	C	CS
14	Flexible markets and advanced electricity trading	C	C	C	CS	C	CS
15	Accelerated carbon removal in line with high quantitative targets	C	C	C	CS	C	S
16	The accelerated elimination of traditional power generation sources	C	C	C	CS	C	CS
17	Digital participation in energy management	CS	CS	C	CS	C	CS
18	Advanced integration between energy and smart urban planning	CS	CS	C	CS	CS	S

Note: S = supportive; CS = conditional supportive; C = constrained. These evaluations indicate the impact of each dimension on the applicability of the element within the Iraqi context, and do not represent a general assessment of the performance of the Iraqi energy sector.

Table A2. Evidential basis, critical constraints, and final classification of energy transformation elements

No.	Energy Transformation Element	Quantitative or Contextual Basis Used	Critical Constraints Identified	Final Classification	Justification for the Classification
1	Improving energy efficiency	Growing demand for electricity, increased consumption by buildings and facilities, and continued pressure on generation capacity and the grid.	Weak enforcement of efficiency standards, limited energy auditing, incentives, and accessible financing.	Highly adaptable	It addresses a direct Iraqi need, and can begin with building, equipment, lighting, and energy auditing standards without waiting for the

					restructuring of the electricity market or the complete rebuilding of the grid.
2	Reducing losses in electricity transmission and distribution	Limited electricity available to consumers, aging parts of the grid, and high technical and non-technical losses	The need to upgrade the network and meters, and improve maintenance, collection, and monitoring.	Highly adaptable	It increases the amount of available electricity without the need to add similar generating capacity, and addresses a direct flaw in the performance of the existing system. It is based on high solar potential and responds to a direct need to diversify sources of supply and reduce reliance on private generation. It can be initiated with rooftop applications and limited projects, with gradual expansion linked to improved connectivity, protection, financing and maintenance. It can begin by raising awareness and managing the workloads of public institutions, then gradually move to a digital response to demand after improving measurement and data. They are an enabling condition for improving billing, monitoring losses, managing demand, and supporting distributed generation.
3	Distributed solar energy	The decline in the contribution of renewable energy compared to the rise in solar potential, and the continued reliance on local and private generation solutions.	Weak connectivity, protection, and control arrangements; limited funding and maintenance; and unclear relationship between the product and the network.	Highly adaptable	They can be implemented initially in selected areas or sectors and then scaled up after assessing cost, reliability, data protection, and operational capabilities. It is a prerequisite and an early step for implementing all other transformation elements. It does not depend on the completion of grid or market reforms before implementation. It also contributes to improved institutional coordination, clarified responsibilities, and enhanced implementation, monitoring, and evaluation capabilities.
4	Electricity demand management	Growing demand for electricity, a persistent supply-demand gap, and weak control over consumption patterns.	Lack of data and measurement, weak incentive tariffs, and limited demand response programs.	Highly adaptable	
5	Smart measurement and data systems	Limited accuracy of consumption data, and the need to improve billing, planning, network monitoring, and losses.	The cost of digital infrastructure, poor connectivity in some areas, data protection and privacy, and a lack of expertise.	Highly adaptable	
6	Building institutional and technical capacities	Disparities in capabilities, weak institutional coordination, and a lack of clarity regarding some responsibilities and procedures.	Lack of specialized expertise, fragmentation of responsibilities, and weak follow-up and implementation.	Highly adaptable	

7	Pilot storage projects	Fluctuations in supply, the need to support solar systems, improve resilience, and reduce reliance on generators.	High cost, limited operational and maintenance experience, and unclear most viable use cases.	Highly adaptable	It can support supply stability, enhance the performance of solar systems, and reduce reliance on generators in priority facilities and areas. It is preferable to begin with a limited pilot project to assess economic and operational feasibility and maintenance requirements before scaling up. They provide a localized solution for under-supply areas, critical facilities, and residential complexes, allowing for the integration of renewable generation, storage, and load management within a defined scope. Pilot projects can be initiated and then scaled up based on performance results and the clarity of ownership, interconnection, financing, and maintenance arrangements.
8	Microgrids	The need for local solutions for under-supplied areas and vital facilities, and the presence of social expertise in decentralized generation.	The absence of clear standards for operation, ownership, connectivity, distribution of responsibilities, financing, and maintenance.	Highly adaptable	The incentive function is suitable for Iraq, but its mechanism needs to be designed gradually to match financial capacity and performance measurement.
9	Incentive tariff for renewable energy production	The need to encourage households, institutions, and investors to produce renewable electricity.	Incomplete procurement, tariff, and connection procedures; weak measurement and collection; and the risk of unsustainable financial obligations.	It requires a major reform	Incentives should be directed towards priority projects and linked to measurable results, while avoiding open-ended financial commitments. The social function is appropriate, but the ownership model needs legal and financial simplification and testing in limited local projects.
10	Financial and market incentives	Limited investment in renewable energy, high entry costs, and financial risks.	Inadequate financing, an unstable investment environment, and limited guarantees and transparency.	It requires a major reform	Limited implementation is possible, but widespread expansion requires significant technical, regulatory, and financial upgrades.
11	Community or cooperative ownership of energy	The potential to increase participation, share benefits, and enhance acceptance, given the existing social reliance on local energy solutions.	The absence of clear legal and financial frameworks for ownership, profit and risk sharing, and weak institutional trust and expertise.	It requires a major reform	This does not represent a realistic starting point before reforming the network, billing, and
12	Large-scale decentralized generation	Increased need for supply diversification and the existence of unregulated private generation systems.	Limited network capacity for bidirectional flows, weak protection, control, measurement, and regulation of producer responsibilities.	It requires a major reform	
13	Extensive liberalization of the electricity market	The potential to improve competition, attract investment, and increase pricing	Weak market maturity, measurement, collection, and	Related to the specificities of the German context or to	

		efficiency in the long term.	control; unclear separation of institutional roles; and risks to fairness and ability to pay.	the requirements of an advanced energy system.	regulation, building an effective regulatory body, and protecting consumers.
14	Flexible markets and advanced electricity trading	The ability to manage supply, demand, and prices, and integrate diverse sources in advanced stages.	Lack of digital and market infrastructure, and weaknesses in real-time measurement, forecasting, oversight, and independent institutions.	Related to the specificities of the German context or to the requirements of an advanced energy system.	It depends on a technical, regulatory, and market structure that is not currently available.
15	Accelerated carbon removal in line with high quantitative targets	High reliance on fossil fuels, limited contribution of renewable energy, and pressure to reduce emissions.	The persistent supply and demand gap, the lack of reliable alternatives, and the limited network, funding, and institutional capacity.	Related to the specificities of the German context or to the requirements of an advanced energy system.	The principle of emissions reduction is appropriate, but Germany's targets, ratios, and timetables are not directly transferable.
16	The accelerated elimination of traditional power generation sources	High reliance on oil and gas for electricity generation, versus the inadequacy of current renewable alternatives	Supply security risks, weak storage and network infrastructure, high demand, and the economy's reliance on fossil fuels.	Related to the specificities of the German context or to the requirements of an advanced energy system	Reliance on traditional sources should be gradually reduced while reliable alternatives are provided.
17	Digital participation in energy management	The need to improve transparency, consumer engagement, and demand management, along with a relative expansion in the use of digital technologies.	The digital divide, unequal access, weak data protection, trust, and digital skills.	It requires a major reform	It can be adopted gradually, but non-digital alternatives must be provided, data protected, and the exclusion of low-income and low-skilled groups prevented. Energy requirements can be integrated into new urban plans and projects, but achieving advanced integration requires extensive institutional coordination, shared databases, and more sophisticated local technical and regulatory capabilities.
18	Advanced integration between energy and smart urban planning	Opportunities exist to integrate solar energy, building efficiency, transportation, and data into urban planning, despite current limited institutional application.	Weak coordination between the energy, planning, transport and communications sectors, and a lack of databases and local authorities.	It requires a major reform	